SRE STUDIUM PRZYPADKU

AZURE ARC HYBRID CLOUD LAB

Czesci 1-10: K8s, Azure Arc, Terraform, CI/CD, Ingress TLS, HPA, NetworkPolicy, RBAC, GitOps, Trivy, Defender, cert-manager, KEDA, Helm, OPA/Gatekeeper, Chaos Mesh, Linkerd

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Data: Czerwiec 2026 (Czesci 1-10)

GitHub: github.com/buth11/sre-homelab-azure-arc

Spis tresci

1–11. Kontekst, Infrastruktura, FinOps, Azure Arc, Observability, KQL, LB, Prometheus, Terraform, Policy, GitHub

12. Czesc 2: Service Principal, Remote State, CI/CD

13. Czesc 3: Ingress TLS, HPA, NetworkPolicy, RBAC

14. Postmortem POST-001

15. Wyniki i wnioski (Czesci 1-3)

16. Troubleshooting Log (10 incydentow)

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26. Wyniki i wnioski (Czesci 1-10)

27. Troubleshooting Log (17 incydentow)

Rozdzial 1: Kontekst i architektura

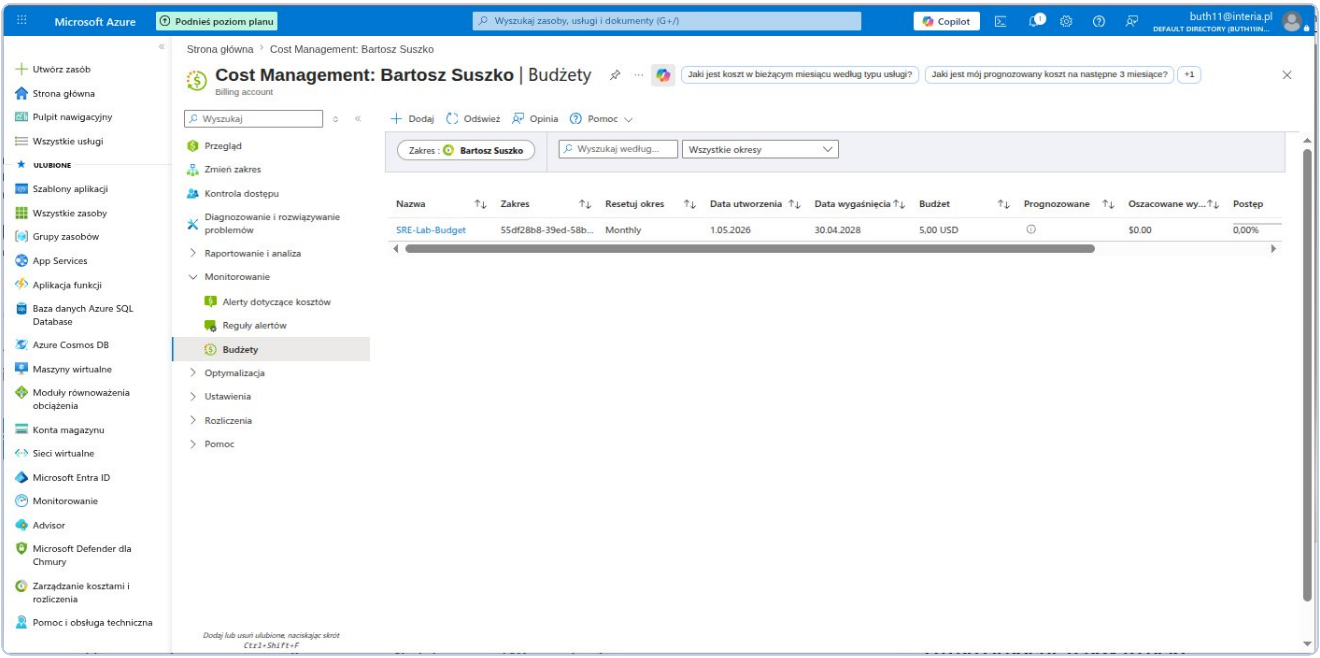
Projekt demonstruje budowe produkcyjnego srodowiska hybrydowego na GMKtec EVO-X2 (128 GB RAM) laczacego Kubernetes z Microsoft Azure przez Azure Arc. Trzy tygodnie praktycznych cwiczen z dokumentacja kazdego kroku i 10 udokumentowanych incydentow.

Warstwa	Technologia	Zastosowanie
Kubernetes	K3s v1.35.4+k3s1	Klaster 3 wezly
Ingress	Traefik + TLS	HTTPS routing po domenie
Autoscaling	HPA (autoscaling/v2)	Skalowanie CPU 2-8 replik
Network Security	NetworkPolicy	Firewall wewnatrz klastra
Access Control	RBAC + ServiceAccount	Least privilege
Cloud Bridge	Azure Arc	Hybrydowa plaszczyzna kontrolna
Monitoring	Azure Monitor + Prometheus	Observability
IaC	Terraform + Remote State	Infrastructure as Code
CI/CD	Azure DevOps	Pipeline z approval gate

Rozdzial 2: Infrastruktura On-Premise

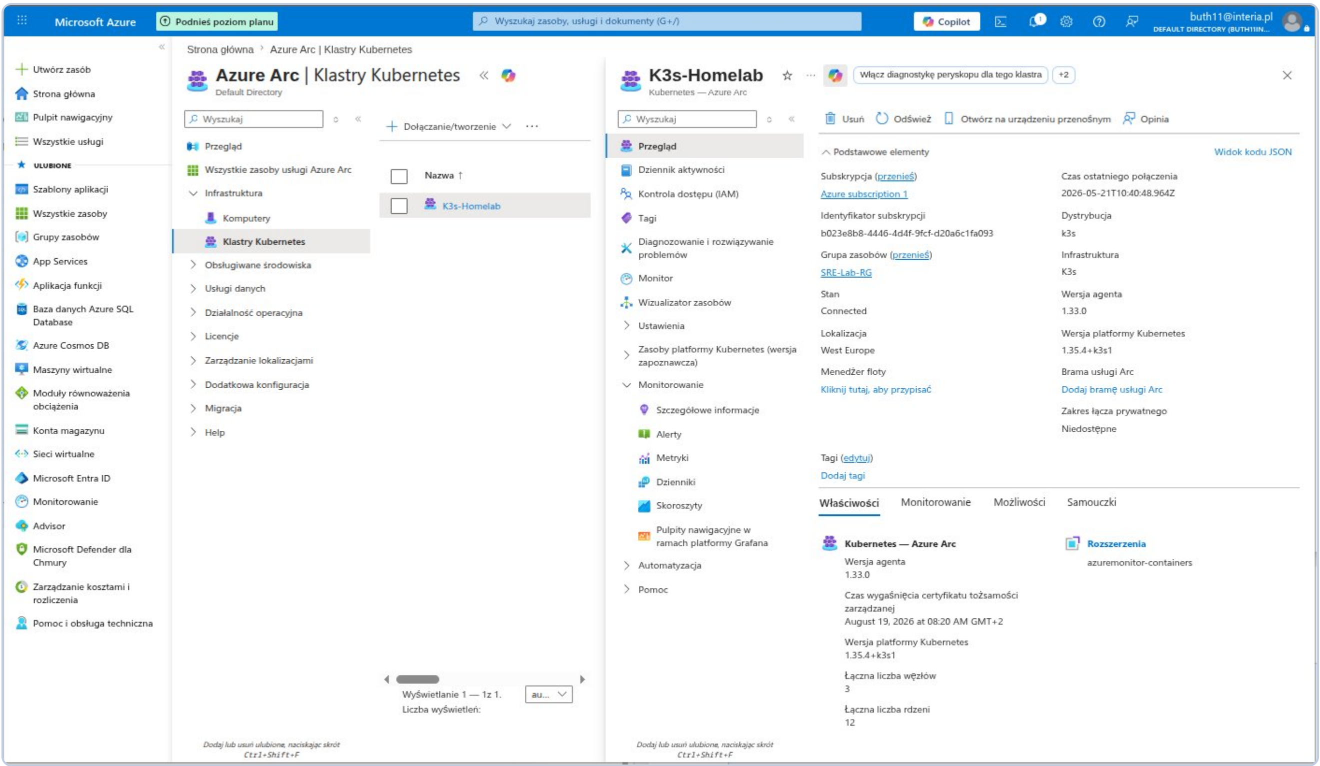
Wezel	IP	CPU	RAM
k3s-master (Control Plane)	192.168.122.10	4 vCPU	8 GB
k3s-worker1	192.168.122.11	4 vCPU	16 GB
k3s-worker2	192.168.122.12	4 vCPU	16 GB

Rozdzial 3: FinOps



Zrzut 1: Azure Cost Management — budzet 5 USD/mies., koszt: 0.00 USD.

Rozdzial 4: Azure Arc



Zrzut 2: Azure Arc — K3s-Homelab Connected, v1.35.4+k3s1, 3 wezly, 12 rdzeni.

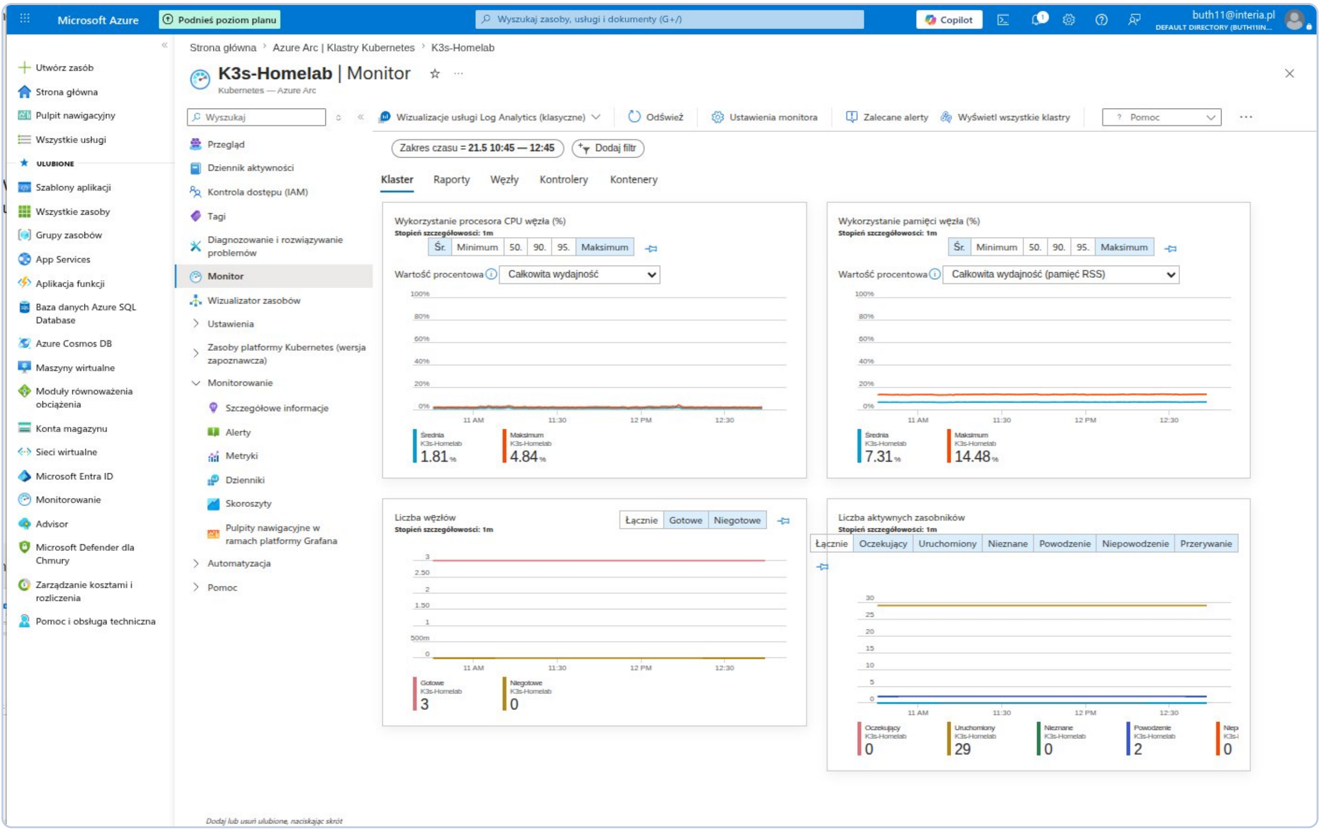
Rozdzial 5: Observability

Desktop : bash | (buth11) 192.168.122.10 | (buth11) 192.168.122.10 | (buth11) 192.168.122.10 | Desktop : bash | (buth11) 192.168.122.10


```
    "principalId": "a3fe4a8c-7a2e-4b99-8de9-2150d617cb76",
    "tenantId": null,
    "type": "SystemAssigned"
  },
  "isSystemExtension": false,
  "name": "azuremonitor-containers",
  "packageUri": null,
  "plan": null,
  "provisioningState": "Succeeded",
  "releaseTrain": "Stable",
  "resourceGroup": "SRE-Lab-RG",
  "scope": {
    "cluster": {
      "releaseNamespace": "azuremonitor-containers"
    },
    "namespace": null
  },
  "statuses": [],
  "systemData": {
    "createdAt": "2026-05-22T07:44:41.936626+00:00",
    "createdBy": null,
    "createdByType": null,
    "lastModifiedAt": "2026-05-22T07:44:41.936626+00:00",
    "lastModifiedBy": null,
    "lastModifiedByType": null
  },
  "type": "Microsoft.KubernetesConfiguration/extensions",
  "version": null
}
```

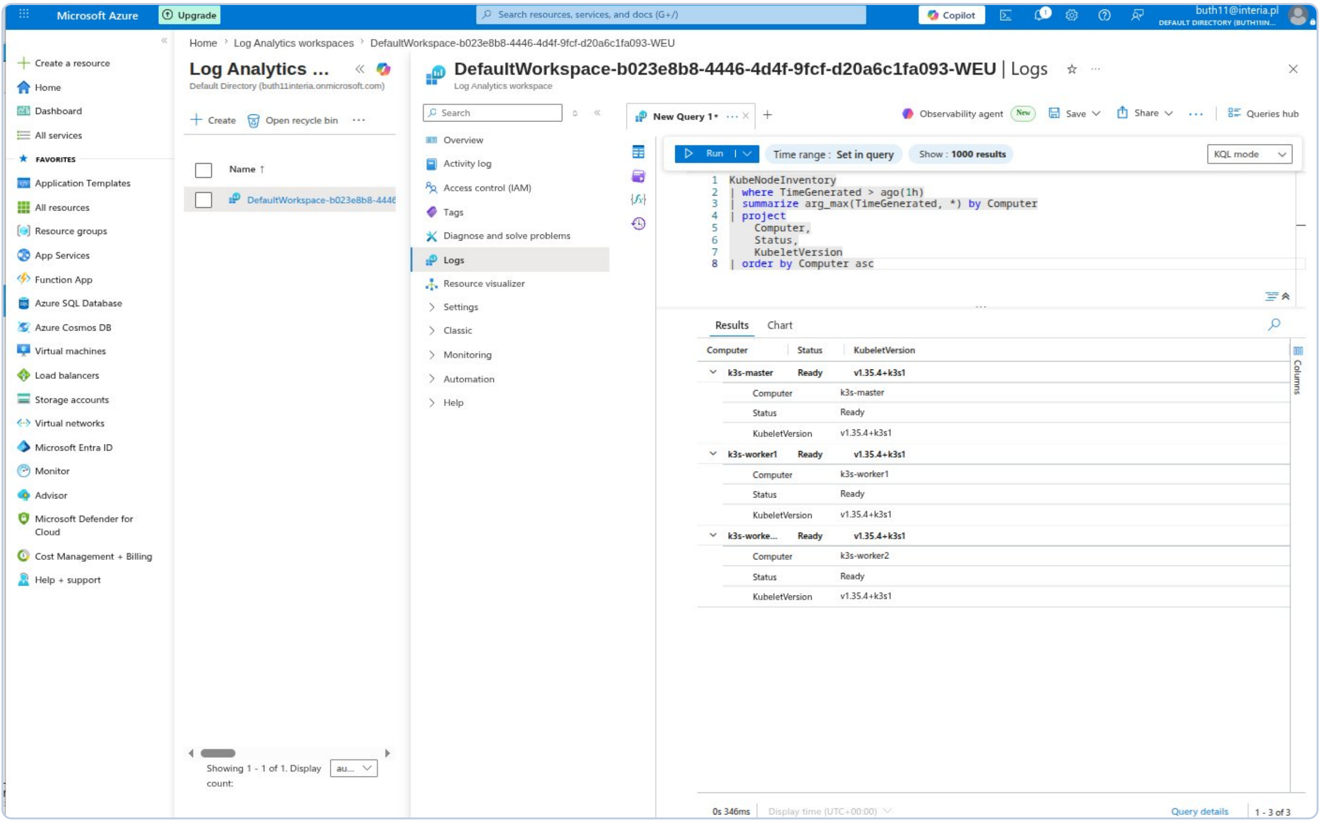

```
buth11@k3s-master:~$ sudo k3s kubectl get pods -n kube-system | grep ama
[sudo] password for buth11:
ama-logs-9m9x9                          3/3     Running     0             37s
ama-logs-9scpr                          3/3     Running     0             37s
ama-logs-rs-65b58c7f96-f8lds            2/2     Running     0             37s
ama-logs-spcdp                          3/3     Running     0             37s
buth11@k3s-master:~$
```

Zrzut 3: Pody AMA Running — 4 agenty, 0 restartow.

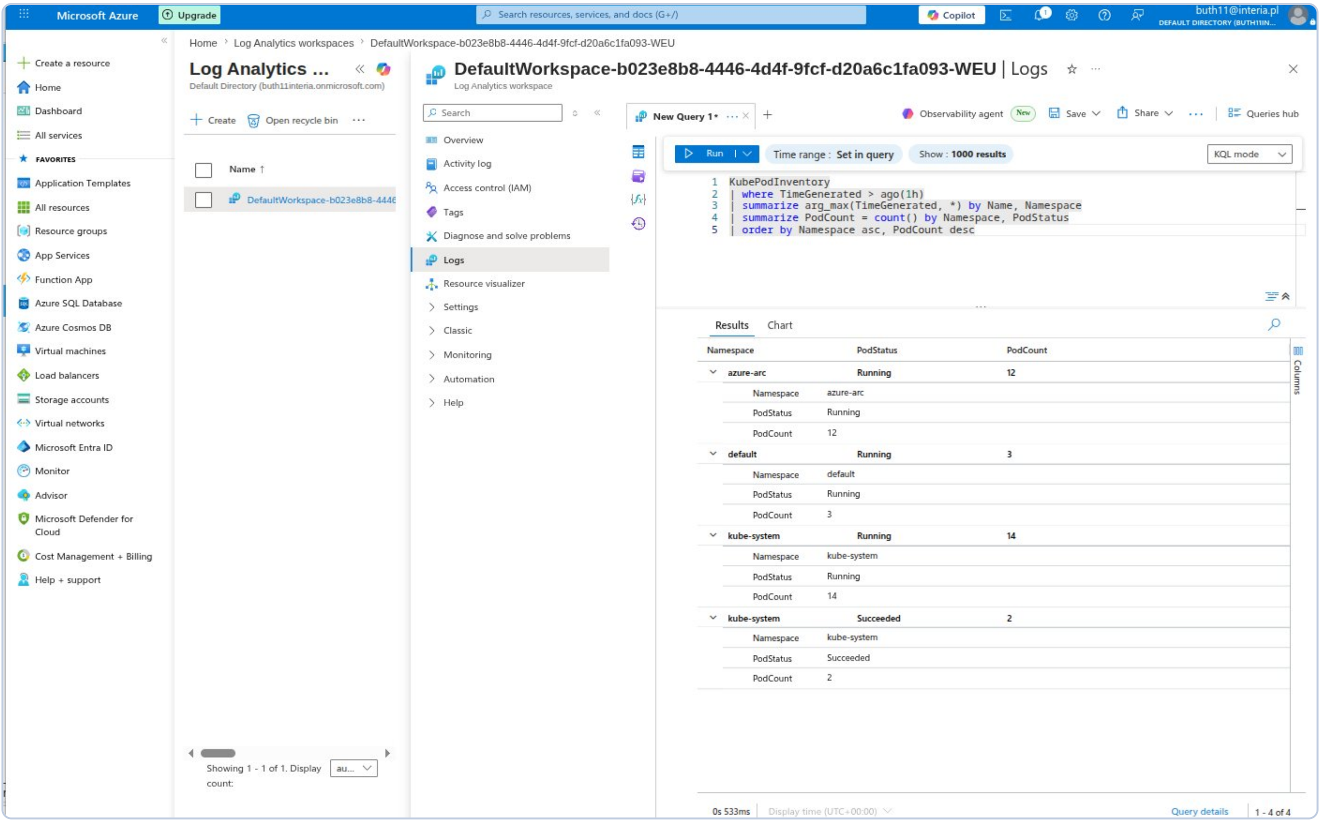


Zrzut 4: Azure Monitor — CPU avg 1.58%, RAM avg 6.27%, 3 wezly Ready, 29 podow.

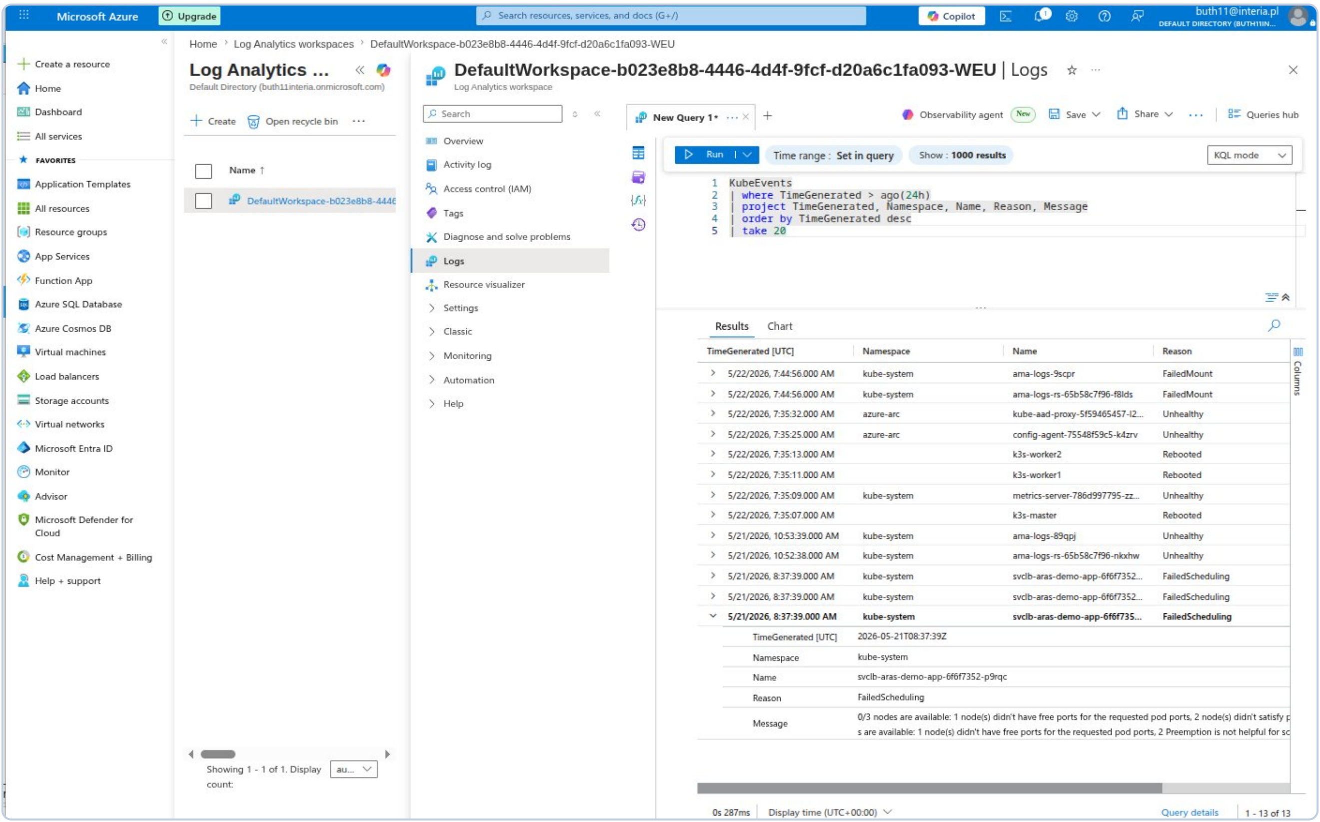
Rozdzial 6: KQL



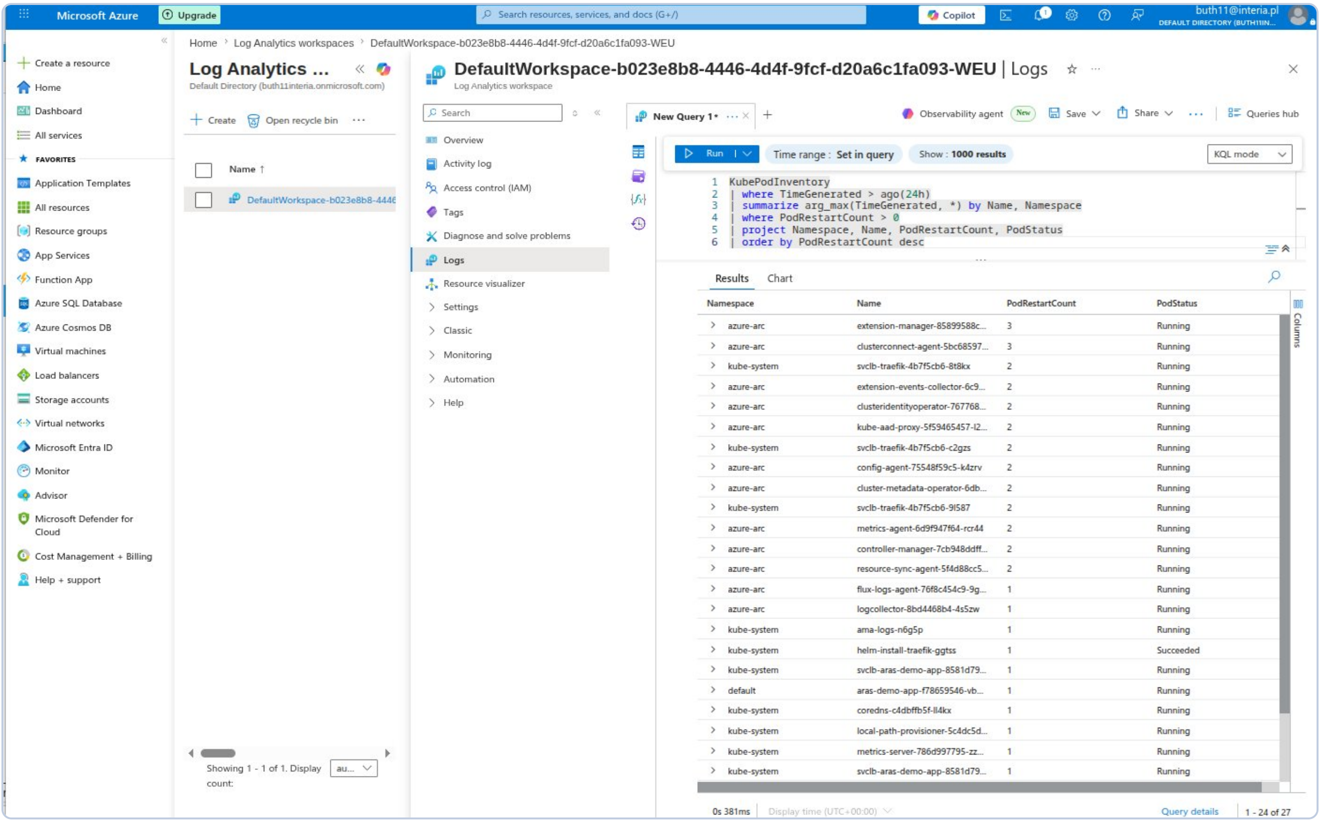
KQL-1: 3 nodes Ready, v1.35.4+k3s1.



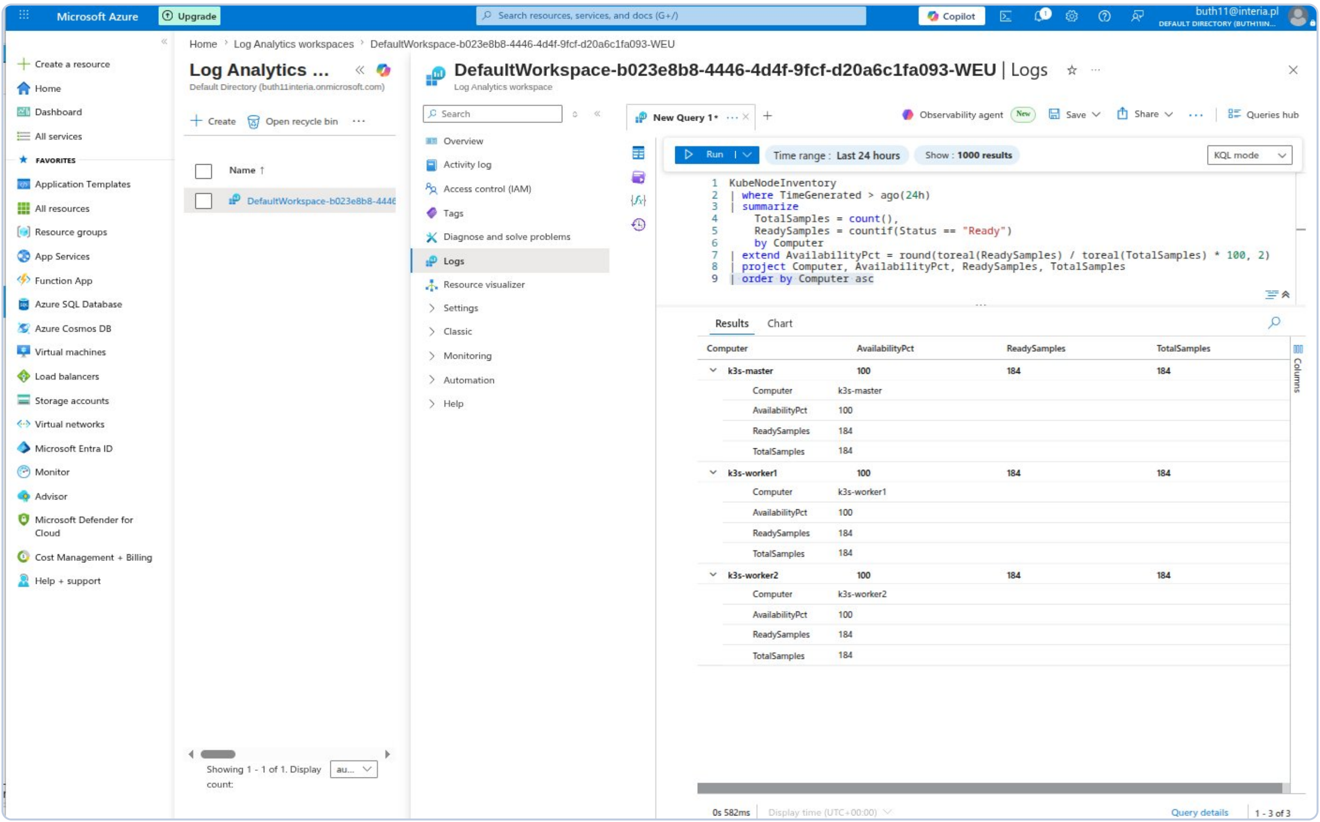
KQL-2: azure-arc 12 Running, default 3 Running, kube-system 14+2.



KQL-3: 13 eventow 24h — Rebooted, Unhealthy (chwilowe), FailedScheduling.



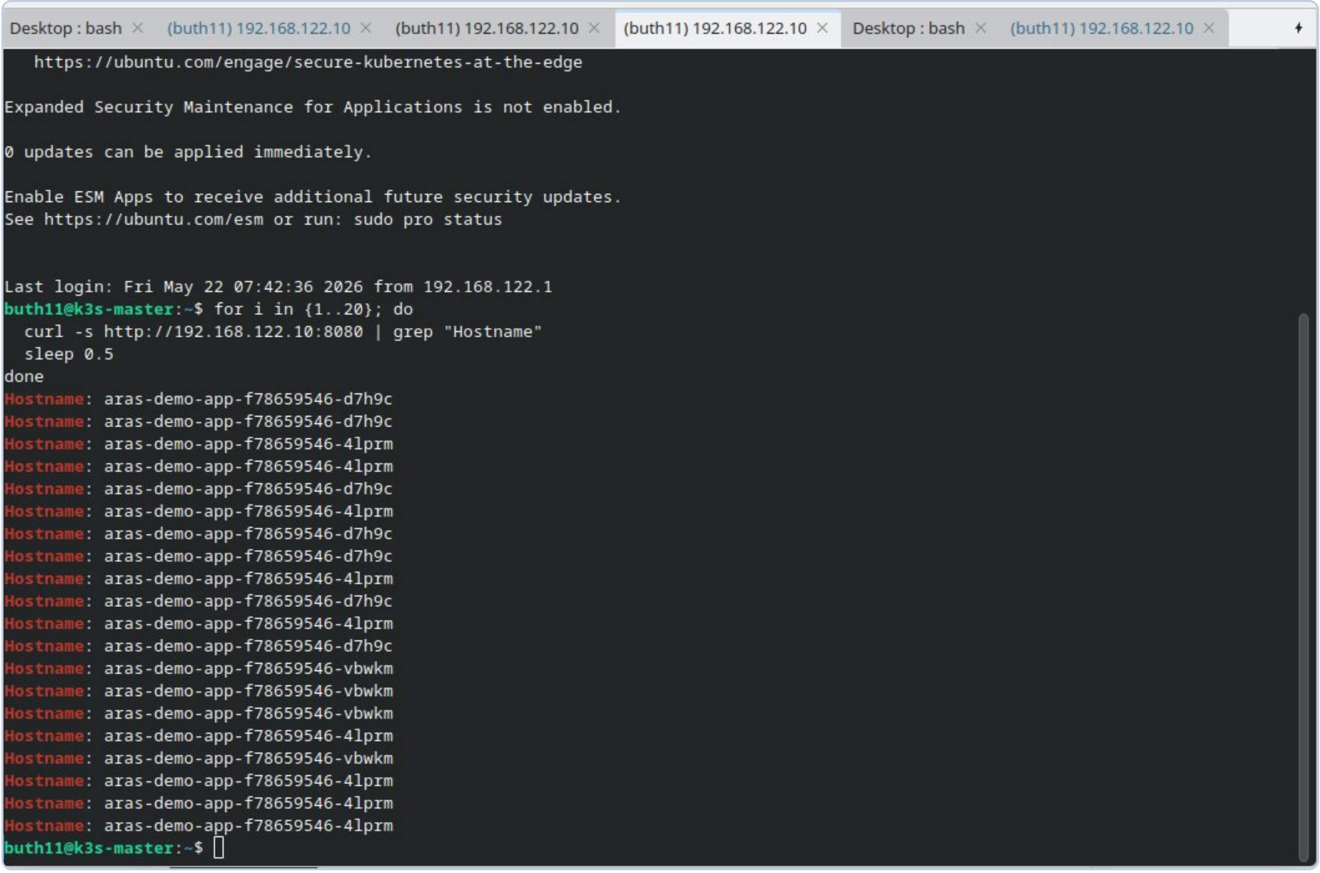
KQL-4: 27 podow z restartami po restarcie VM — wszystkie Running.



KQL-5: SLO 100% — 184/184 probek Ready.

SLO 100% w oknie 24h. Error budget nienaruszony nawet po pelnym restarcie VM.

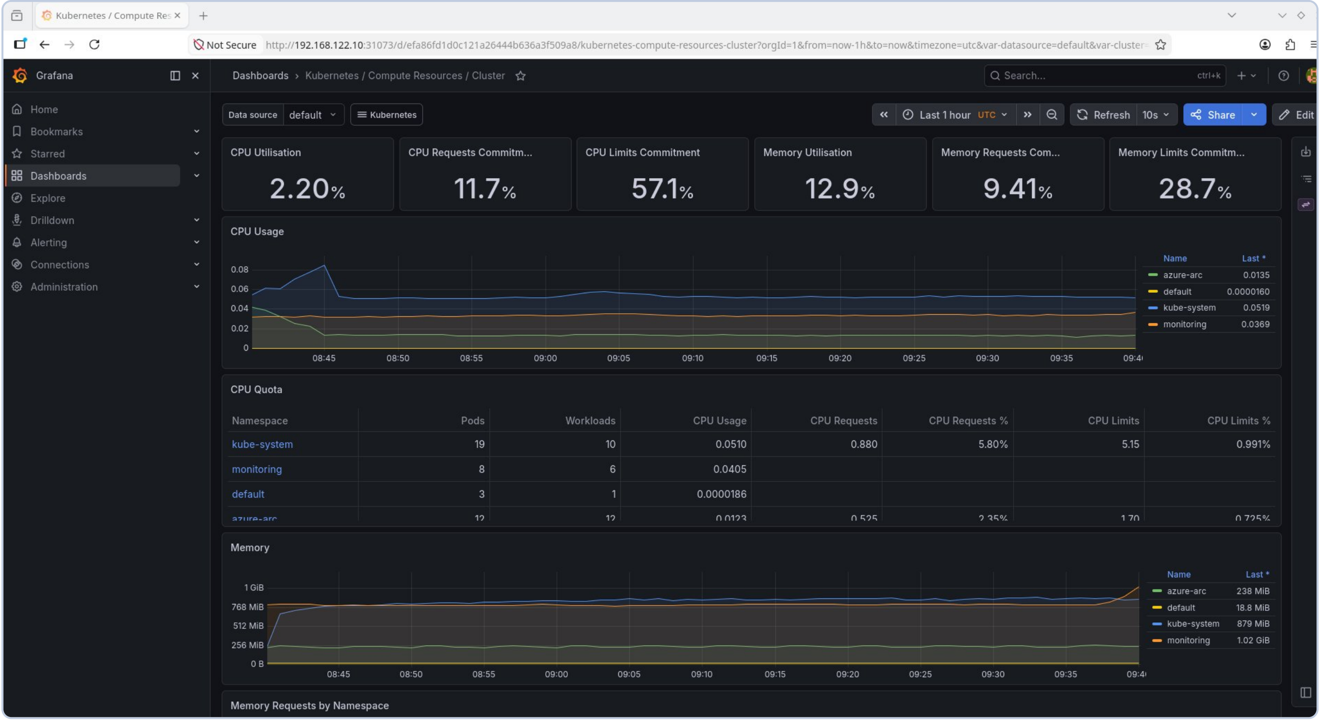
Rozdzial 7: Load Balancing




```
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Fri May 22 07:42:36 2026 from 192.168.122.1
buth11@k3s-master:~$ for i in {1..20}; do
  curl -s http://192.168.122.10:8080 | grep "Hostname"
  sleep 0.5
done
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
buth11@k3s-master:~$
```

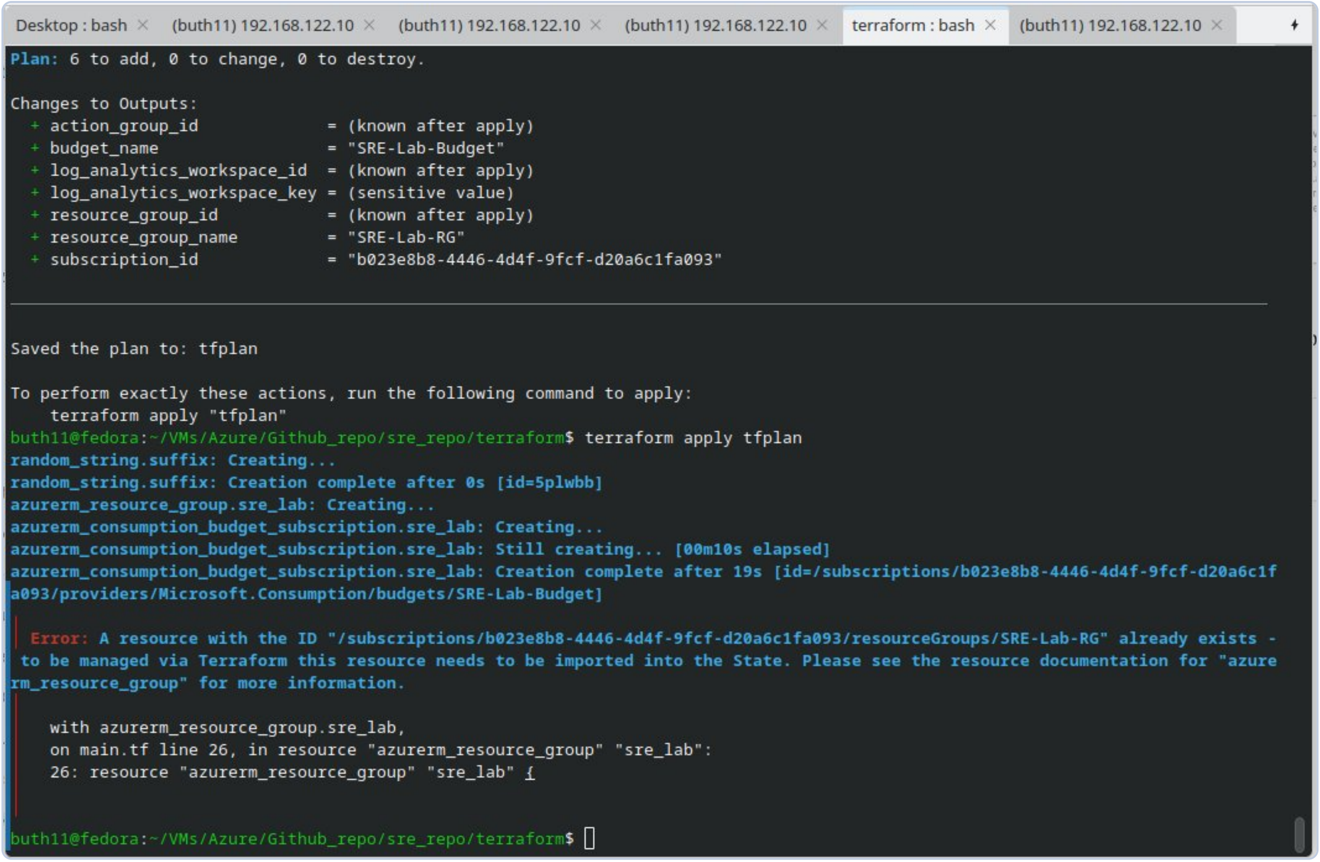
Zrzut 5: Round-robin — 20 requestow na 3 pody, 900 rownolegla bez bledow.

Rozdzial 8: Prometheus & Grafana

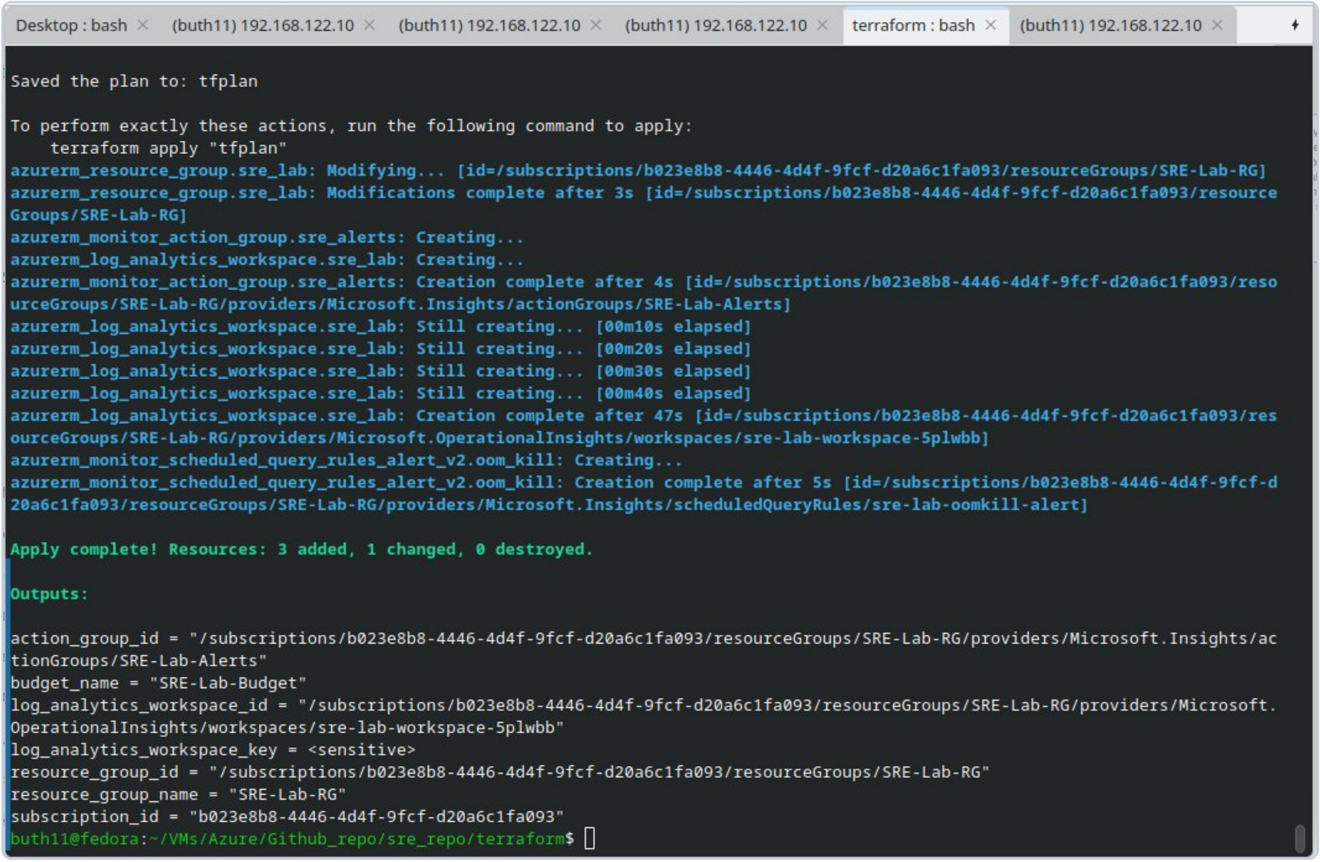


Zrzut 6: Grafana — CPU 2.20%, RAM 12.9%, CPU Quota per namespace.

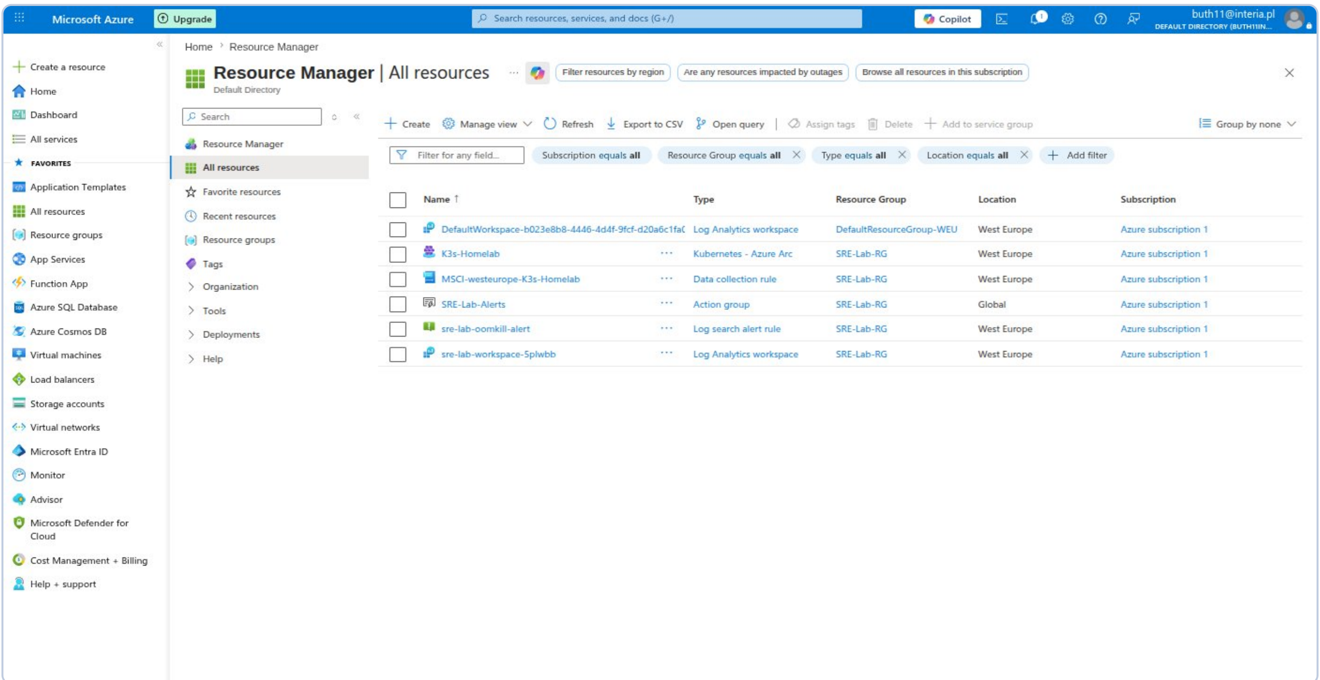
Rozdzial 9: Terraform IaC



> Terraform Plan: 6 zasobow, 0 zmian.

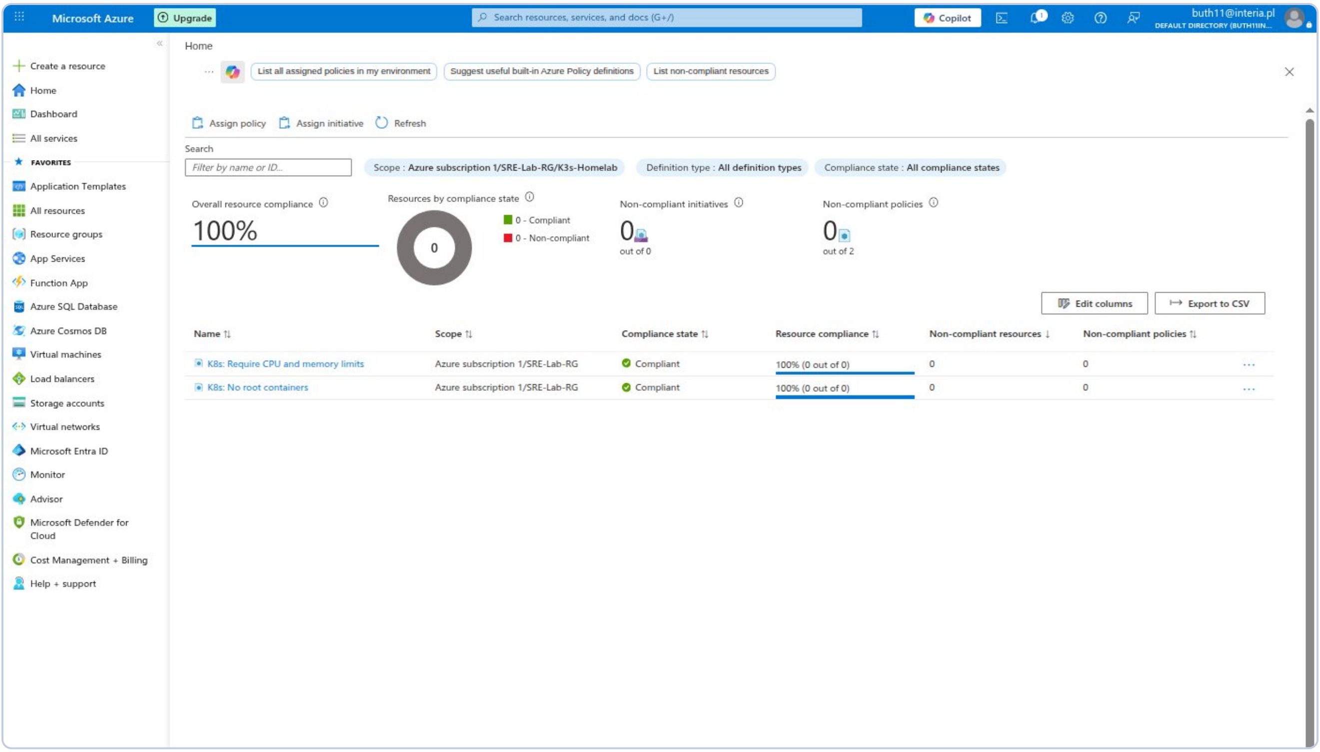


Terraform Apply: 3 dodane, 1 zmieniony.



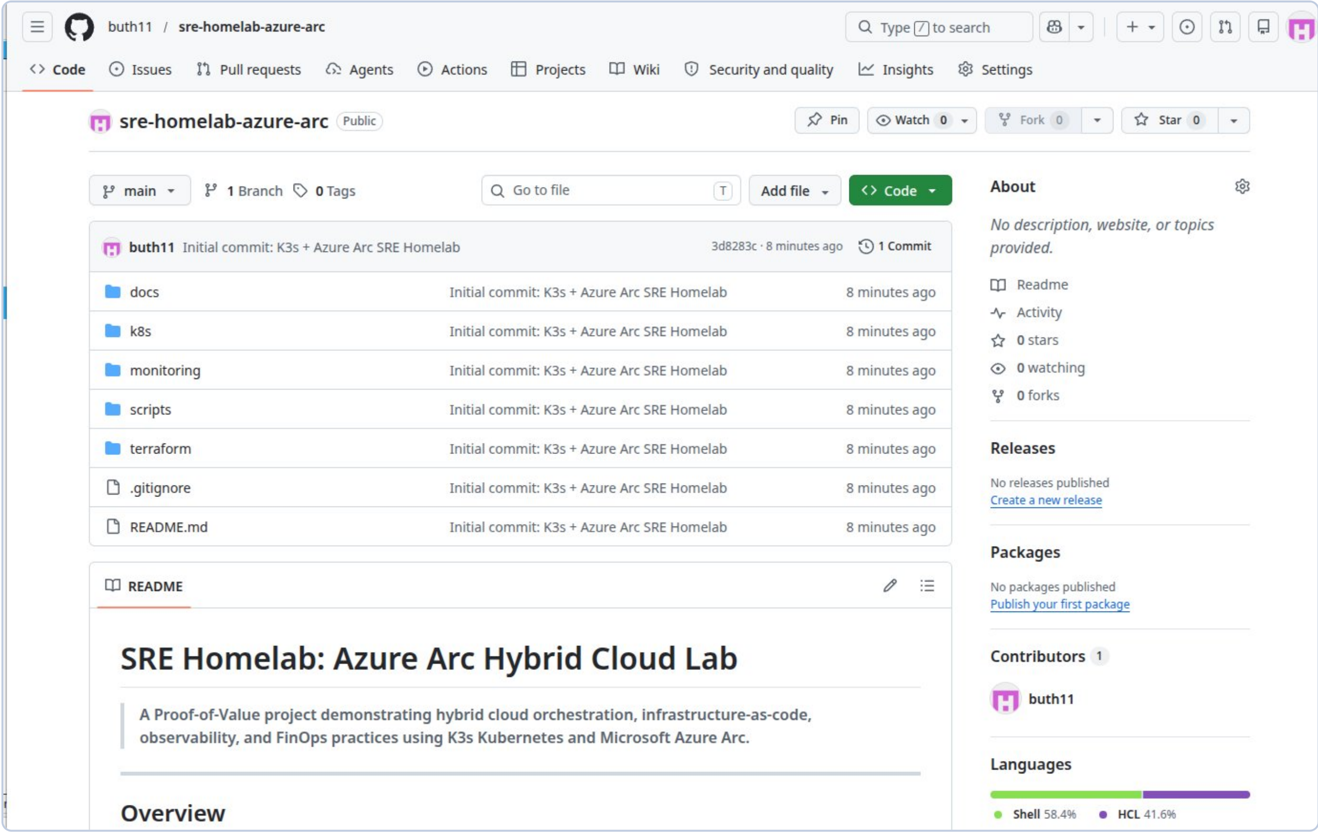
Portal Azure: zasoby Terraform w SRE-Lab-RG.

Rozdzial 10: Azure Policy



Azure Policy: No root containers + CPU/memory limits — oba Compliant.

Rozdzial 11: GitHub

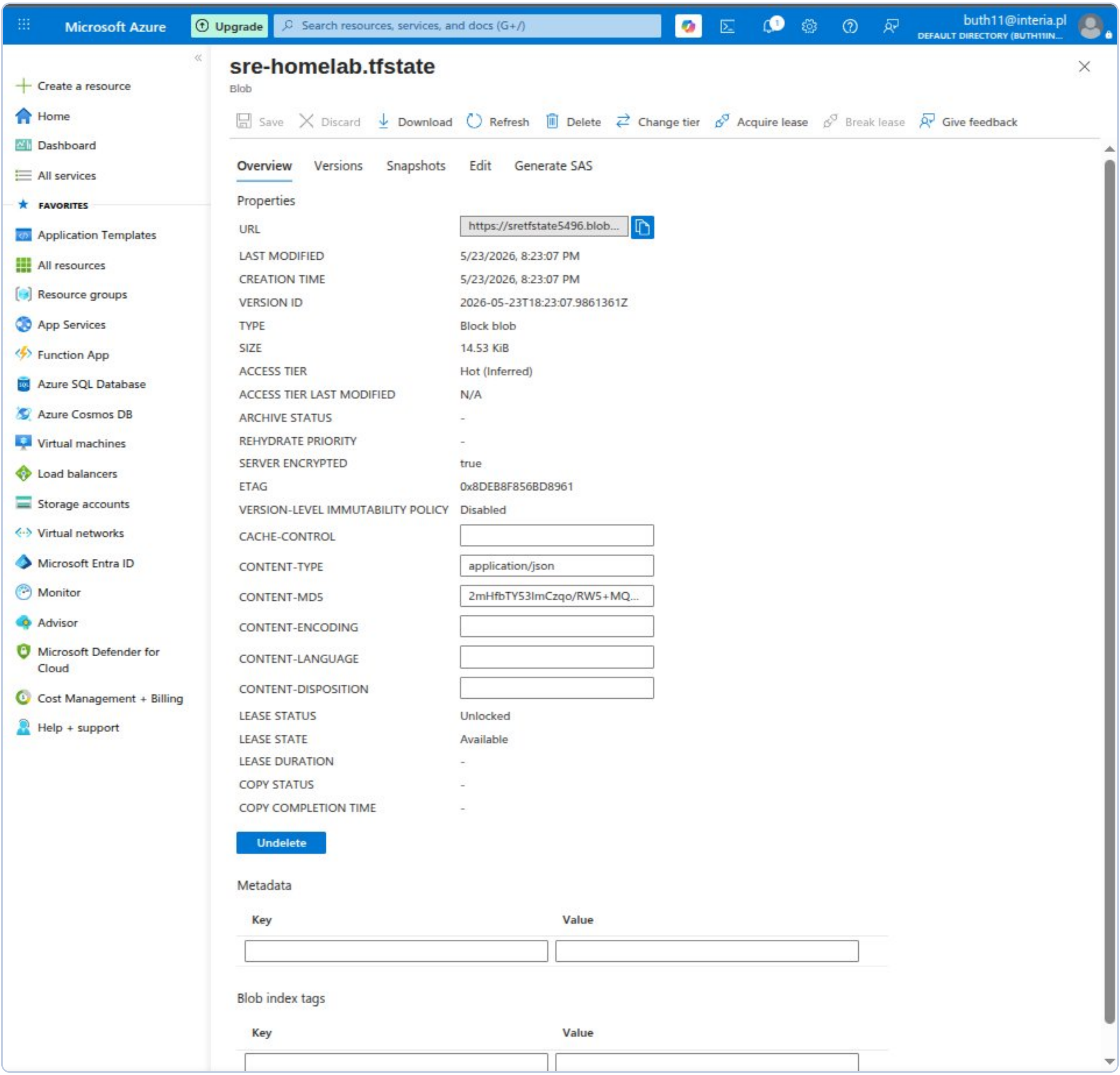


GitHub repo sre-homelab-azure-arc — publiczne, Shell 58.4%, HCL 41.6%.

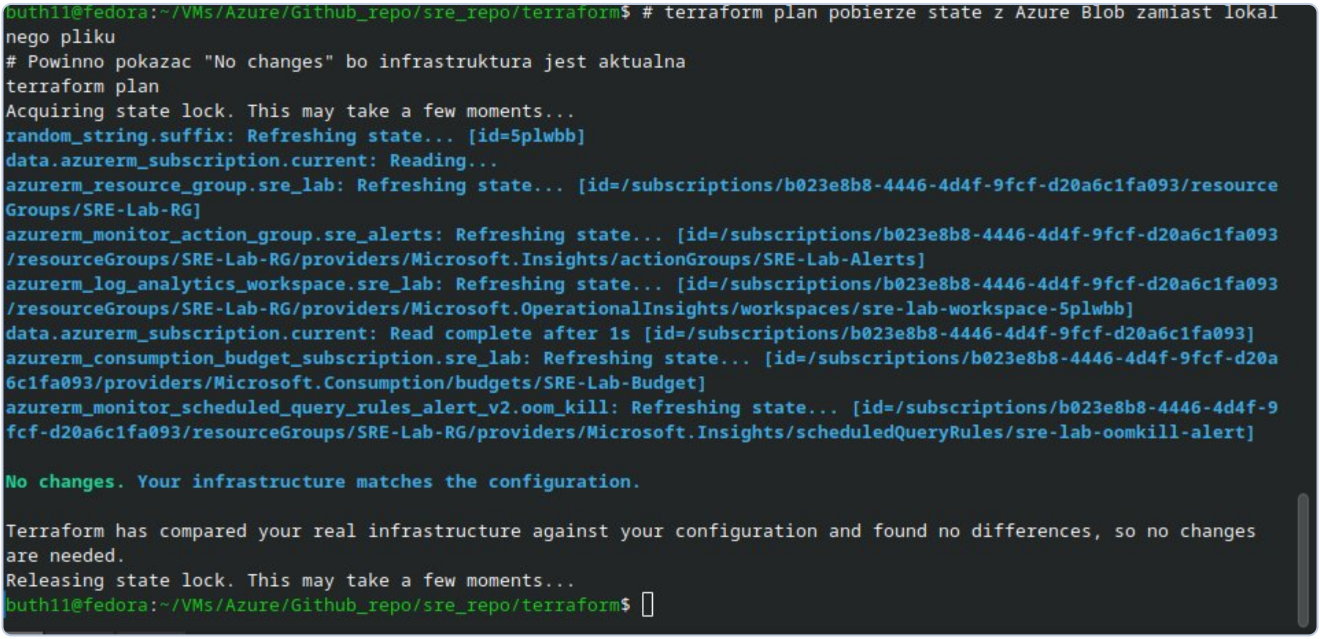
Rozdzial 12: Czesc 2 — Service Principal, Remote State, CI/CD

Remote State — 14.53 KiB, szyfrowanie, wersjonowanie. State lock przy kazdym plan.

Pipeline DevOps: validate + plan (37s) → Approval Gate → apply (49s) = 4m 24s total.

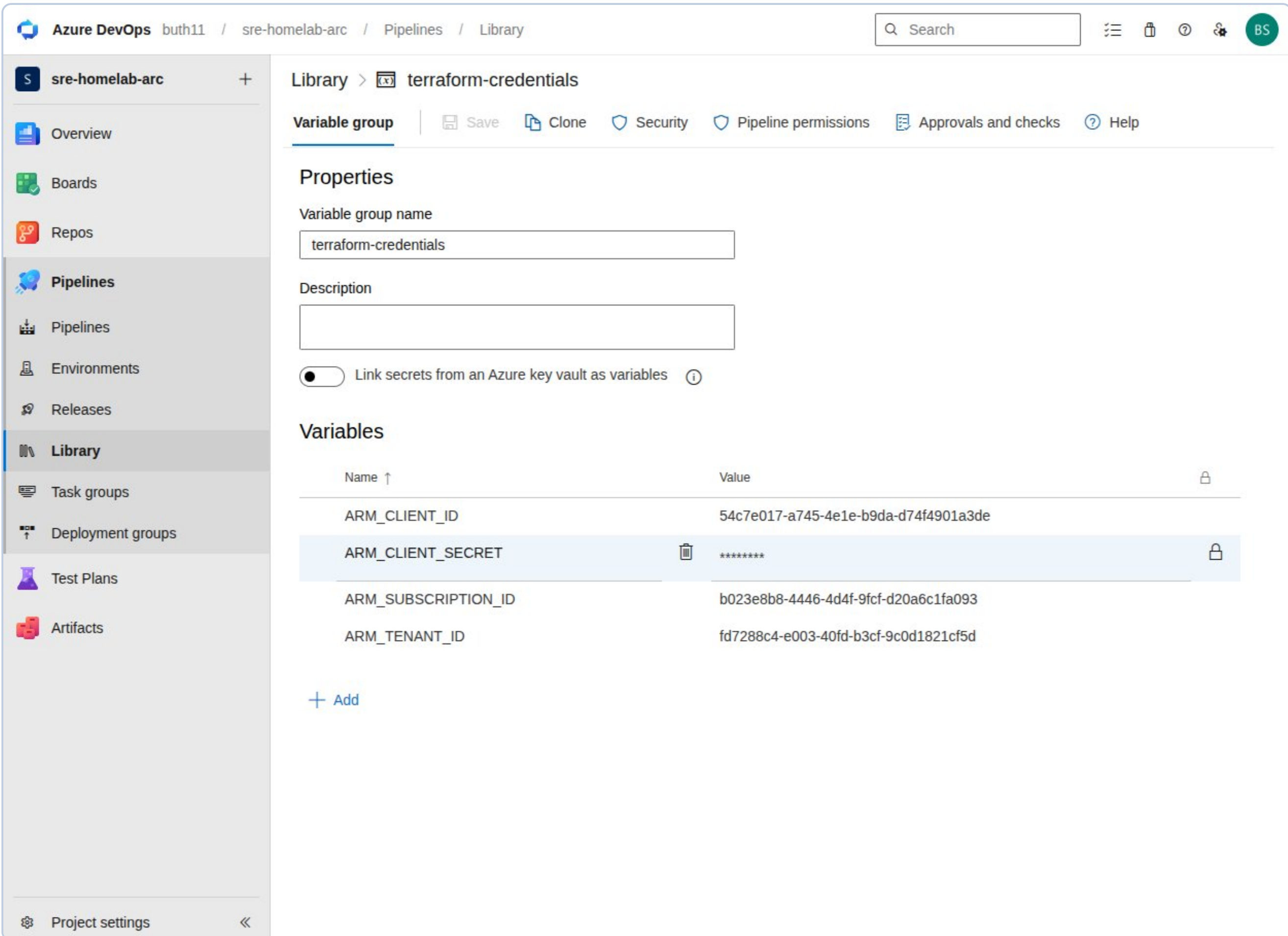


Zrzut W2-2: Portal Azure — blob properties: 14.53 KiB, encrypted, versioning active.

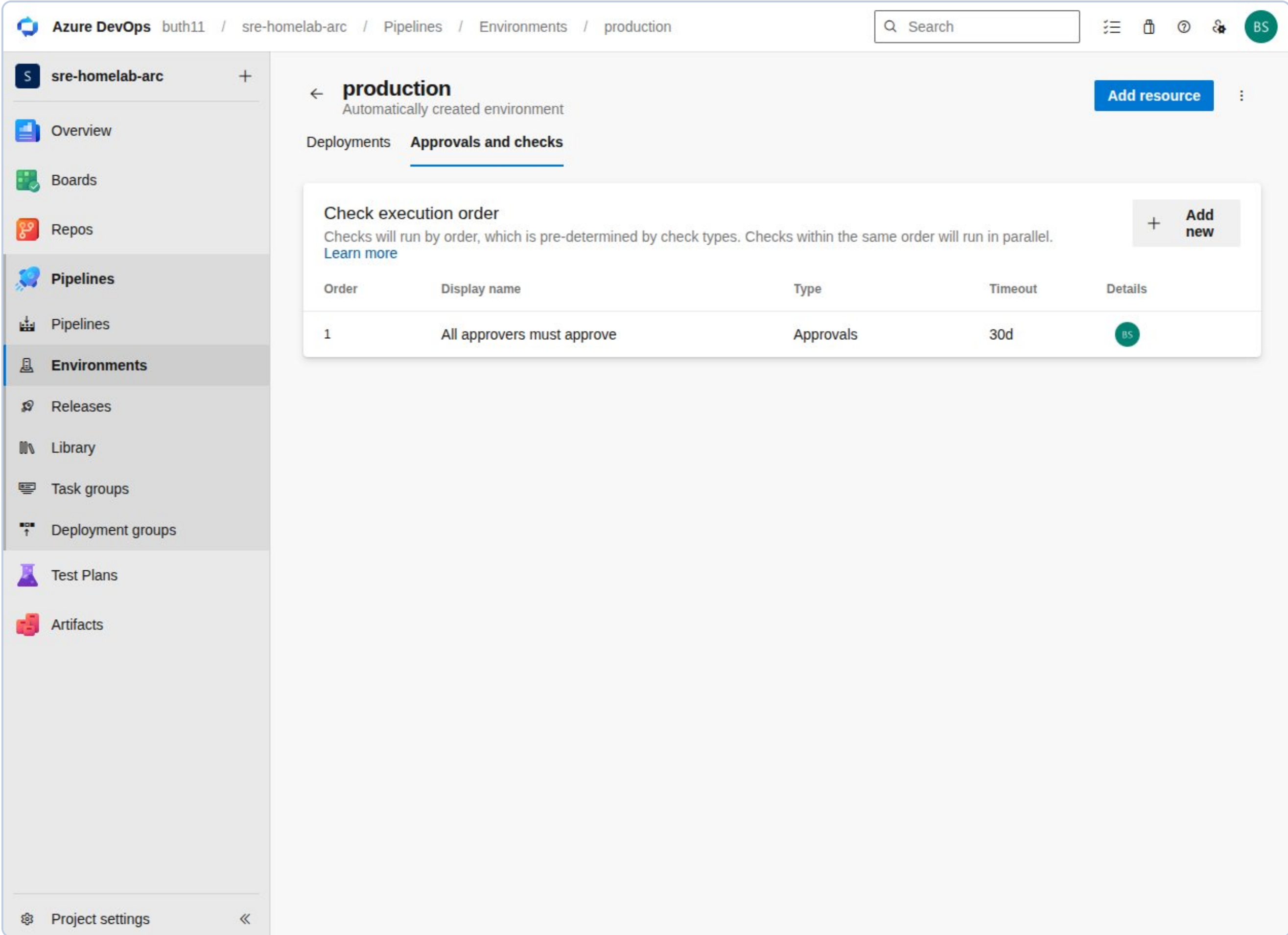


```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # terraform plan pobierze state z Azure Blob zamiast lokal
nego pliku
# Powinno pokazac "No changes" bo infrastruktura jest aktualna
terraform plan
Acquiring state lock. This may take a few moments...
random_string.suffix: Refreshing state... [id=5plwbb]
data.azurerm_subscription.current: Reading...
azurerm_resource_group.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resource
Groups/SRE-Lab-RG]
azurerm_monitor_action_group.sre_alerts: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerm_log_analytics_workspace.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
data.azurerm_subscription.current: Read complete after 1s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093]
azurerm_consumption_budget_subscription.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a
6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9
fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes
are needed.
Releasing state lock. This may take a few moments...
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

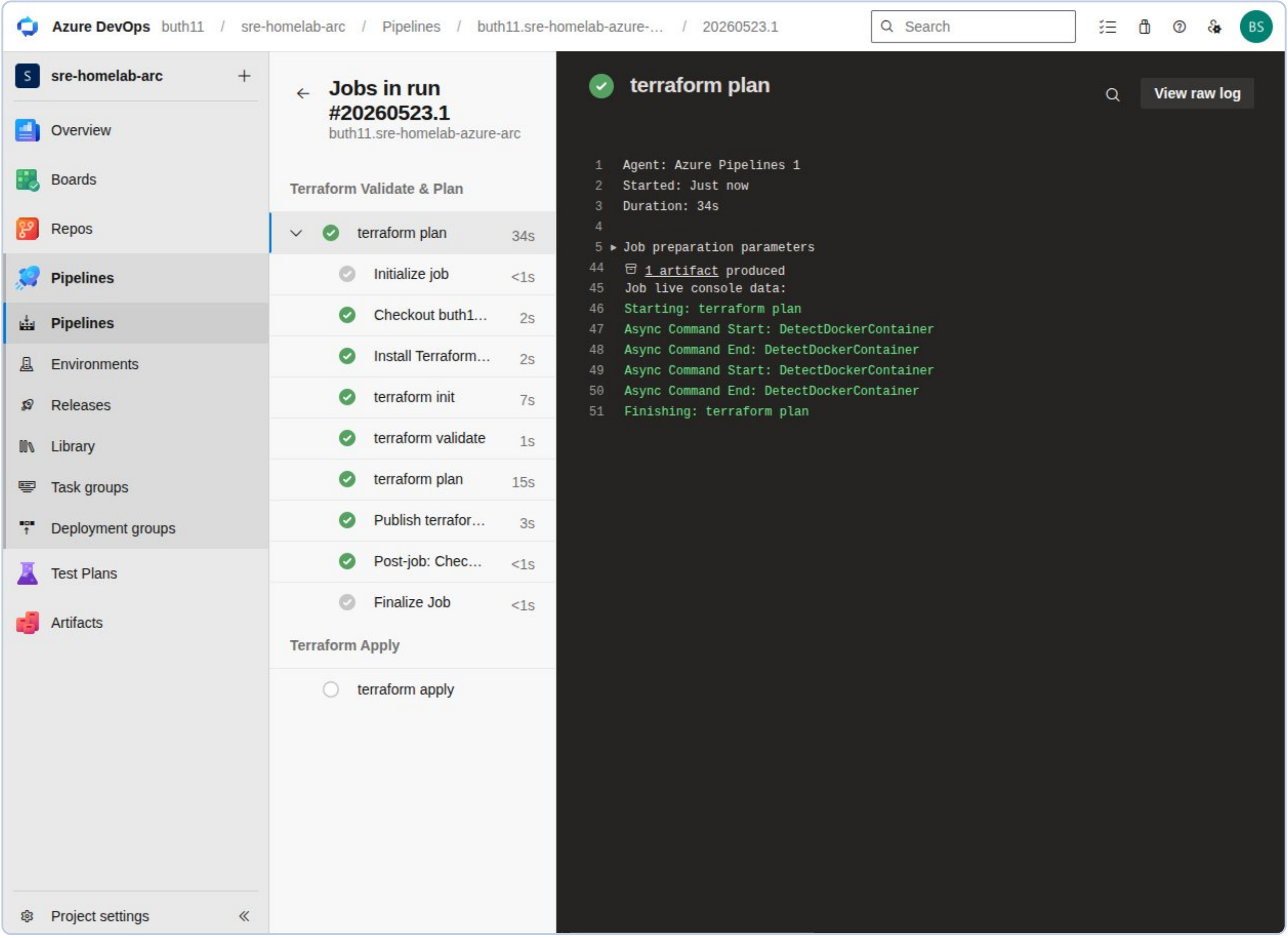
Zrzut W2-3: terraform plan — Acquiring/Releasing state lock. No changes.



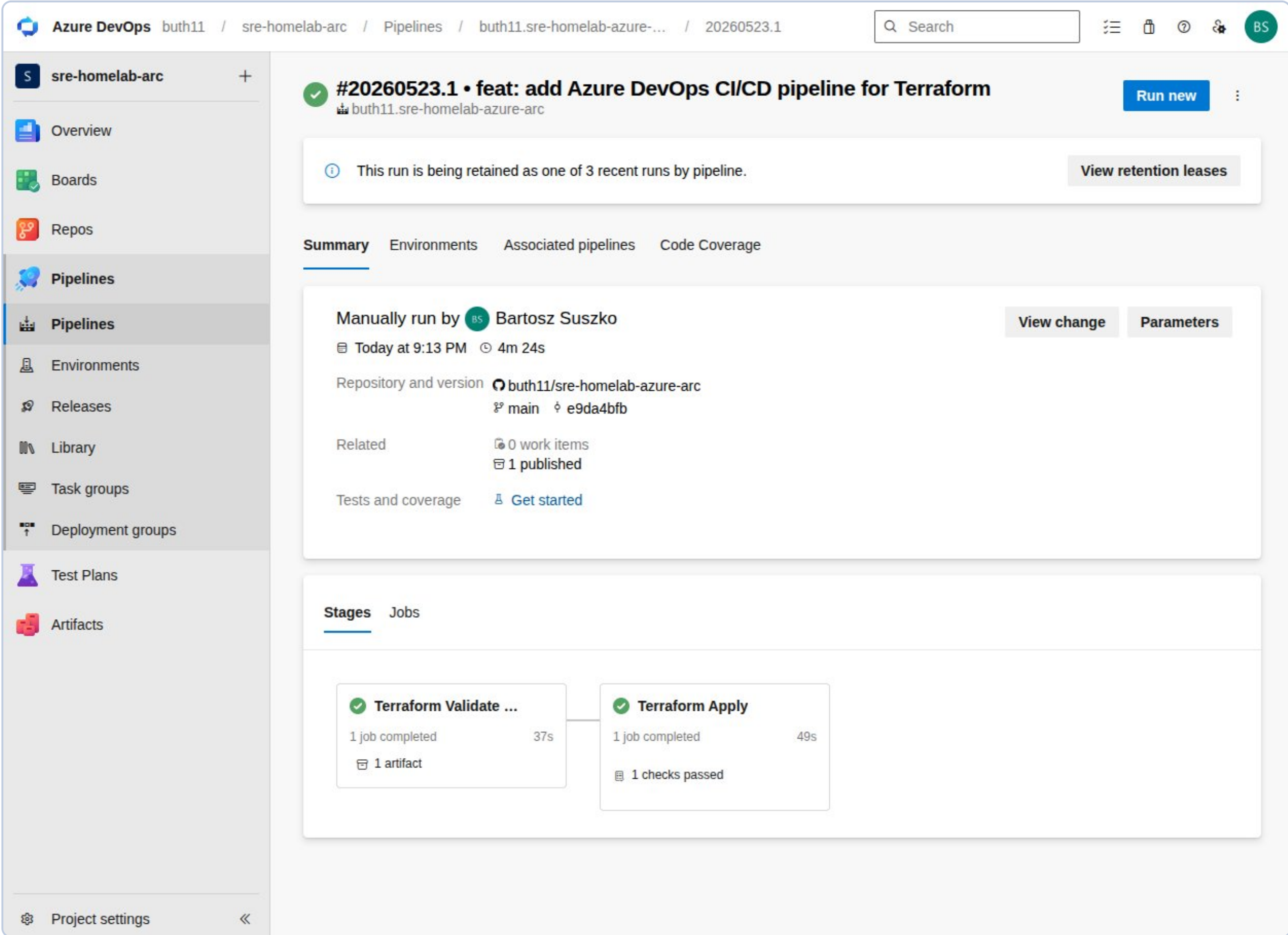
Zrzut W2-4: Azure DevOps Variable Group — ARM_* z ukrytym CLIENT_SECRET.



Zrzut W2-5: Environment production — Approval Gate aktywna.



Zrzut W2-6: Stage 1 — wszystkie kroki zielone, 1 artifact.

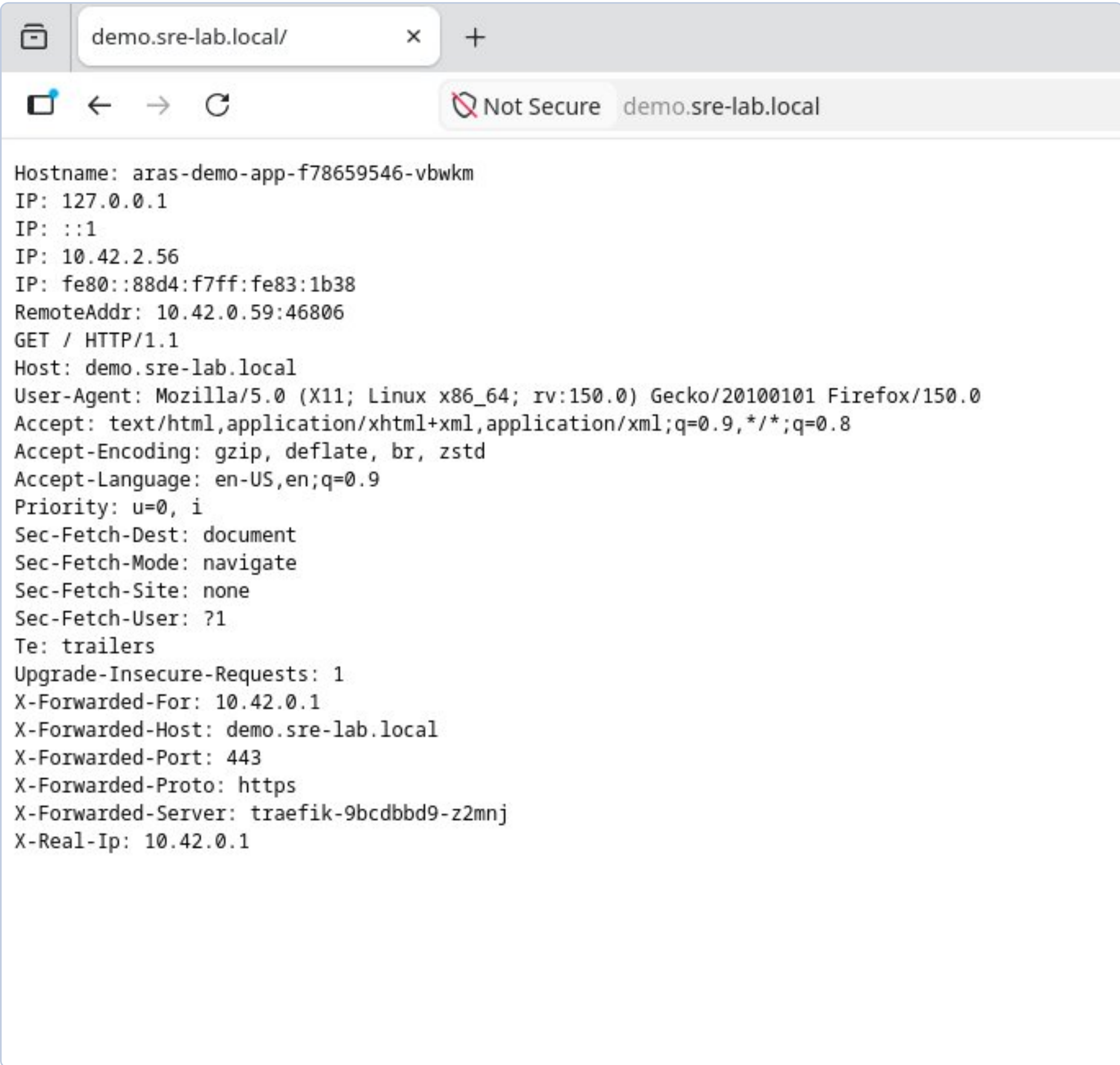


Zrzut W2-7: Pipeline kompletny — oba stage zielone, 4m 24s.

Rozdzial 13: Czesc 3 — Zaawansowane operacje Kubernetes

CZESC 3: Ingress TLS, HPA, NetworkPolicy, RBAC

Traefik HTTPS, self-signed TLS, HPA 2-8 replik (860% CPU), NetworkPolicy deny-all, RBAC least privilege.



> Zrzut W3-1: Przeglądarka Firefox — https://demo.sre-lab.local działa. Widoczne: Hostname poda, X-Forwarded-Proto: https, X-Forwarded-Server: traefik-9bcdbbd9-z2mnj, X-Forwarded-Port: 443.

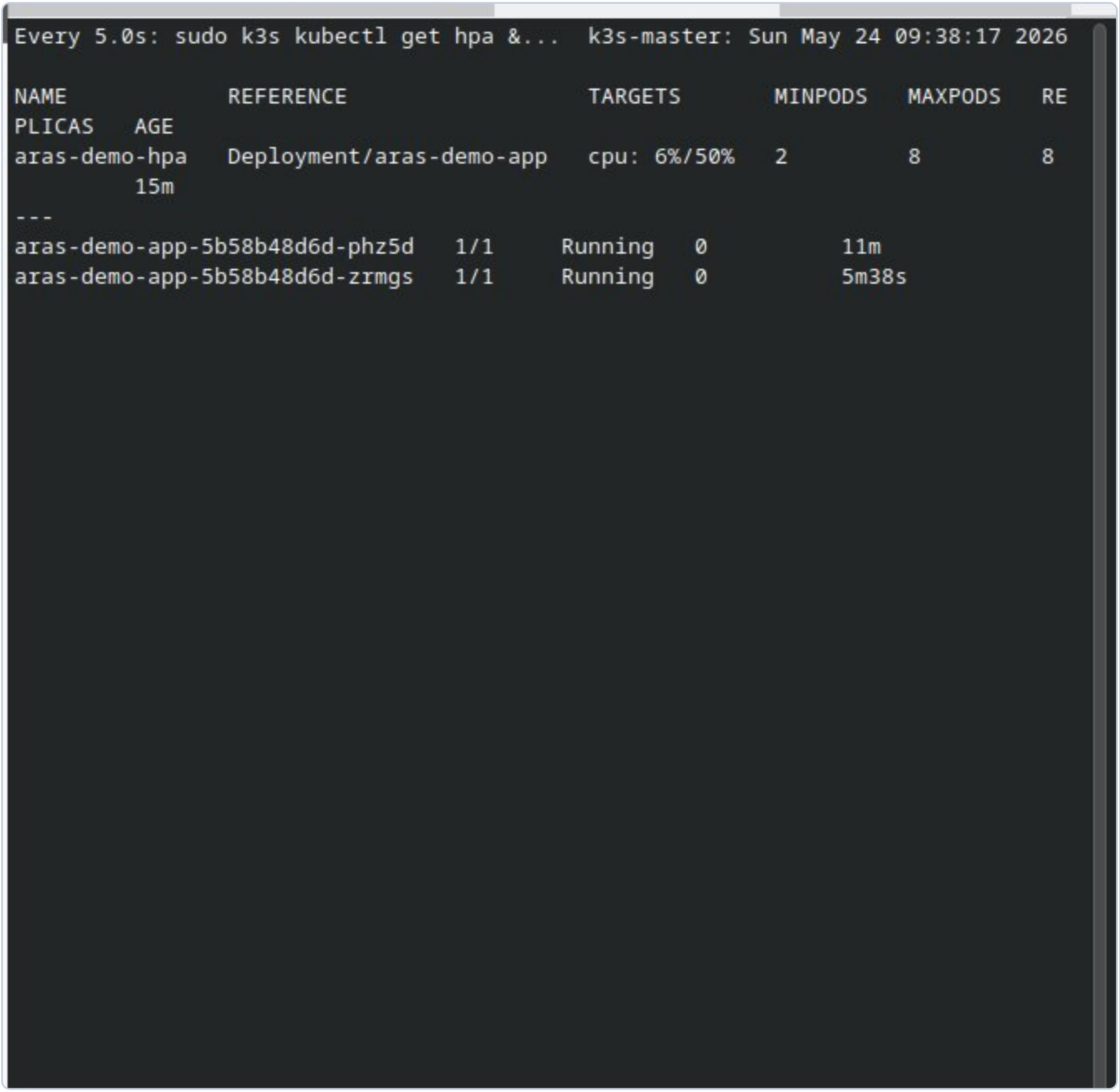
HPA — automatyczne skalowanie na podstawie CPU

Horizontal Pod Autoscaler skonfigurowany dla Deployment aras-demo-app. HPA co 15 sekund odpytuje metrics-server o zuzycie CPU. Gdy srednie CPU Podow przekroczy 50% — dodaje repliki. Gdy spada ponizej 50% przez 5 minut — usuwa.

- Zakres: minReplicas: 2, maxReplicas: 8, targetCPU: 50%
- Test obciazeniowy: ab -n 50000 -c 50 — 1621 req/s przez 30 sekund
- Scale-up: CPU 860% → automatyczne dodanie replik do 8
- Scale-down: CPU 6% po 5 minutach → powrot do 2 replik (minimum)

```
Every 5.0s: sudo k3s kubectl get hpa &...  k3s-master: Sun May 24 09:32:49 2026

NAME             REFERENCE                  TARGETS         MINPODS   MAXPODS
REPLICAS   AGE
aras-demo-hpa   Deployment/aras-demo-app   cpu: 860%/50%   2         8
4          9m55s
---
aras-demo-app-5b58b48d6d-5vwl4   1/1     Running   0          24s
aras-demo-app-5b58b48d6d-7mbln   1/1     Running   0          10s
aras-demo-app-5b58b48d6d-859c7   1/1     Running   0          6m32s
aras-demo-app-5b58b48d6d-96qs4   1/1     Running   0          10s
aras-demo-app-5b58b48d6d-phz5d   1/1     Running   0          6m27s
aras-demo-app-5b58b48d6d-sbv4q   1/1     Running   0          10s
aras-demo-app-5b58b48d6d-xx94p   1/1     Running   0          24s
aras-demo-app-5b58b48d6d-zrmgs   1/1     Running   0          10s
```

Zrzut W3-2: HPA scale-up — cpu: 860%/50%, REPLICAS: 8 (max), 8 podow Running. Automatyczne skalowanie w akcji podczas testu obciazeniowego ab.



```
Every 5.0s: sudo k3s kubectl get hpa &...  k3s-master: Sun May 24 09:38:17 2026

NAME            REFERENCE                  TARGETS       MINPODS    MAXPODS    RE
PLICAS   AGE
aras-demo-hpa   Deployment/aras-demo-app   cpu: 6%/50%   2          8          8
         15m
---
aras-demo-app-5b58b48d6d-phz5d   1/1     Running   0          11m
aras-demo-app-5b58b48d6d-zrmgs   1/1     Running   0          5m38s
```

Zrzut W3-3: HPA scale-down — cpu: 6%/50%, REPLICAS: 2 (min). Po 5 minutach bez ruchu HPA automatycznie usnal 6 zbednych Podow.

NetworkPolicy — firewall wewnatrz klastra

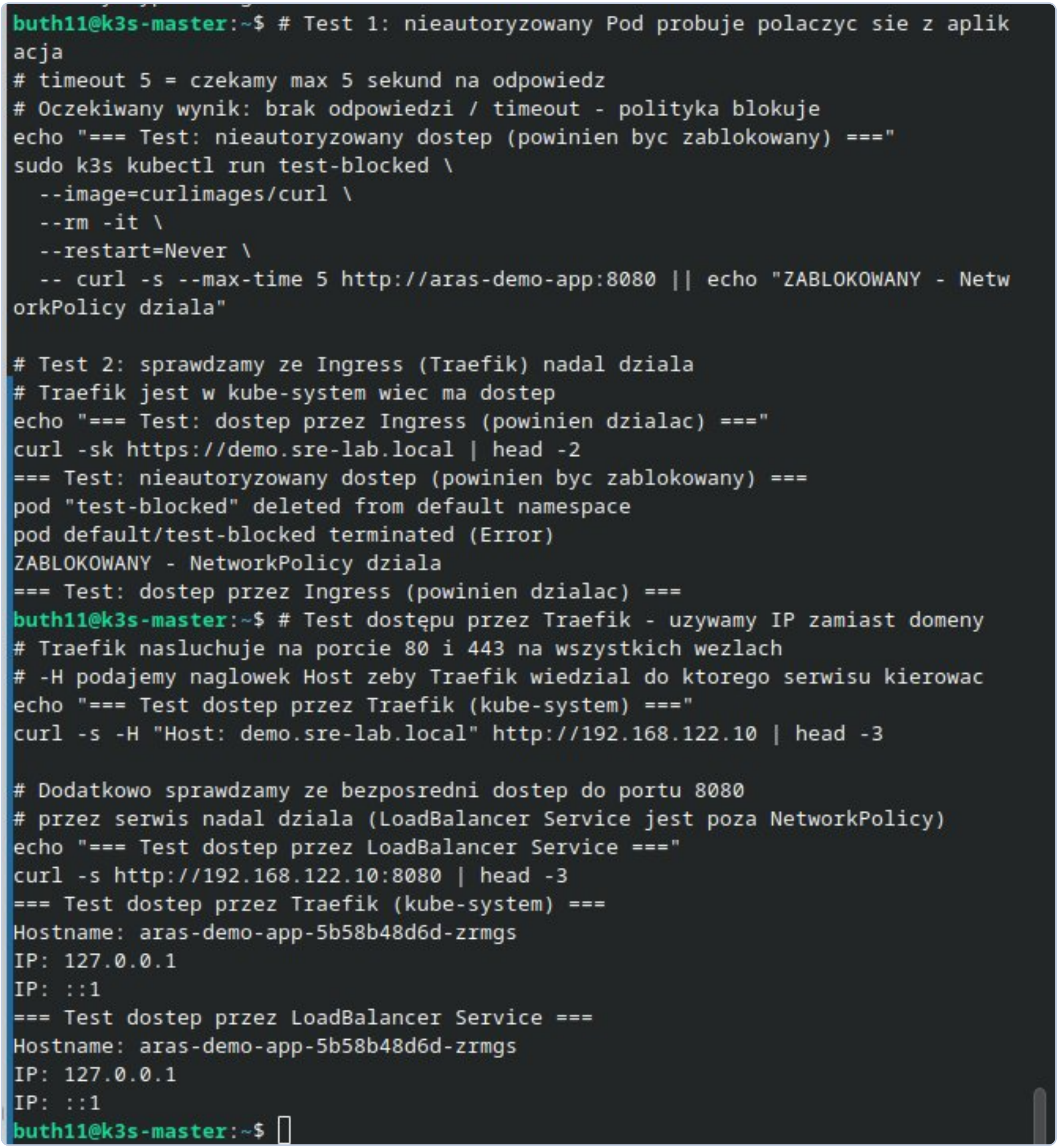
Domyslnie wszystkie Pody w Kubernetes moga sie komunikowac ze wszystkimi. NetworkPolicy definiuje reguly jak firewall — ktory namespace i ktore Pody moga lacyc sie z aplikacja demo. Wdrozono polityke blokujaca caly ruch przychodzacy z wyjatkiem dozwolonych zrodel.

- Zezwolone: kube-system (Traefik), monitoring (Prometheus), repliki aras-demo-app
- Zablokowane: wszystko inne
- Weryfikacja: nieautoryzowany Pod — timeout/blokada. Traefik — odpowiedz aplikacji

RBAC — least privilege dla aplikacji

Role-Based Access Control — kazda aplikacja powinna miec wlasne konto (ServiceAccount) z dokladnie tymi uprawnieniami ktorych potrzebuje. Zasada least privilege: jesli aplikacja nie potrzebuje kasowac Podow, nie powinna moc tego robic nawet jesli zostanie przejeta przez atakujacego.

- ServiceAccount: aras-demo-sa (zamiast wspolnego default)
- Role: get/list/watch na Podach i ConfigMapach — nic wiecej
- Weryfikacja: can-i list pods = yes, can-i delete pods = no



```
buth11@k3s-master:~$ # Test 1: nieautoryzowany Pod probuje polaczyc sie z aplik
acja
# timeout 5 = czekamy max 5 sekund na odpowiedz
# Oczekiwany wynik: brak odpowiedzi / timeout - polityka blokuje
echo "=== Test: nieautoryzowany dostep (powinien byc zablokowany) ==="
sudo k3s kubectl run test-blocked \
  --image=curlimages/curl \
  --rm -it \
  --restart=Never \
  -- curl -s --max-time 5 http://aras-demo-app:8080 || echo "ZABLOKOWANY - Netw
orkPolicy dziala"

# Test 2: sprawdzamy ze Ingress (Traefik) nadal dziala
# Traefik jest w kube-system wiec ma dostep
echo "=== Test: dostep przez Ingress (powinien dzialac) ==="
curl -sk https://demo.sre-lab.local | head -2
=== Test: nieautoryzowany dostep (powinien byc zablokowany) ===
pod "test-blocked" deleted from default namespace
pod default/test-blocked terminated (Error)
ZABLOKOWANY - NetworkPolicy dziala
=== Test: dostep przez Ingress (powinien dzialac) ===
buth11@k3s-master:~$ # Test dostępu przez Traefik - uzywamy IP zamiast domeny
# Traefik nasluchuje na porcie 80 i 443 na wszystkich wezlach
# -H podajemy naglowek Host zeby Traefik wiedzial do ktorego serwisu kierowac
echo "=== Test dostep przez Traefik (kube-system) ==="
curl -s -H "Host: demo.sre-lab.local" http://192.168.122.10 | head -3

# Dodatkowo sprawdzamy ze bezposredni dostep do portu 8080
# przez serwis nadal dziala (LoadBalancer Service jest poza NetworkPolicy)
echo "=== Test dostep przez LoadBalancer Service ==="
curl -s http://192.168.122.10:8080 | head -3
=== Test dostep przez Traefik (kube-system) ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
=== Test dostep przez LoadBalancer Service ===
Hostname: aras-demo-app-5b58b48d6d-zrmgs
IP: 127.0.0.1
IP: ::1
buth11@k3s-master:~$
```

Zrzut W3-4: RBAC weryfikacja — ServiceAccount aras-demo-sa: list pods=yes (dozwolone), delete pods=no (zablokowane). Zasada least privilege potwierdzona.

Rozdzial 14: Postmortem POST-001

Pole	Wartosc
Severity	P2
Czas	Ponizej 7 minut
Root Cause	Literowka: ks3-worker1 zamiast k3s-worker1
Status	Rozwiazany — regex validation w skryptach

Rozdzial 15: Wyniki i wnioski (Czesci 1-3)

Metryka	Czesc 1	Czesc 2	Czesc 3
Wezly	3/3 Ready	—	—
SLO	100%	—	—
Koszty	$0.00	$0.00	$0.00
Terraform state	Local	Remote (Blob)	—
CI/CD	—	DevOps 4m24s	—
Ingress TLS	—	—	HTTPS verified
HPA	—	—	3 na 8 na 2
NetworkPolicy	—	—	Zablokowane
RBAC	—	—	Least privilege
Incydenty	6	8	10

Rozdzial 16: Troubleshooting Log

ISSUE-001 KQL: CPUCapacityNanoCores column missing

```
project operator: Failed to resolve scalar expression named CPUCapacityNanoCores
```

Rozwiazanie: KubeNodeInventory | getschema — uproszczono zapytanie do dostepnych kolumn.

ISSUE-002 Azure Policy: MissingPolicyParameter cpuLimit/memoryLimit

```
(MissingPolicyParameter) policy assignment k8s-resource-limits is missing cpuLimit, memoryLimit
```

Rozwiazanie: Dodano --params z cpuLimit: 500m i memoryLimit: 512Mi.

ISSUE-003 Terraform: Resource Group already exists (brownfield)

```
Error: A resource with the ID .../SRE-Lab-RG already exists
```

Rozwiazanie: terraform import azurerm_resource_group.sre_lab /subscriptions/.../SRE-Lab-RG

ISSUE-004 Terraform: Saved plan is stale after import

```
Error: Saved plan is stale — state was changed after plan was created
```

Rozwiazanie: Nowy terraform plan -out=tfplan po imporcie.

ISSUE-005 Container Insights missing after VM restart

Rozwiazanie: az k8s-extension create --name azuremonitor-containers — agenty wrocily w 37s.

ISSUE-006 POST-001: etcd Split-Brain (ks3 vs k3s)

```
FATA[0034] Node password rejected, duplicate hostname
```

Rozwiazanie: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent. Skrypty zaktualizowane o regex validation.

ISSUE-007 Azure DevOps: TerraformInstaller task missing

```
A task is missing: TerraformInstaller
```

Rozwiazanie: Instalacja z Marketplace: ms-devlabs.custom-terraform-tasks.

ISSUE-008 Pipeline: permission for Variable Group

```
This pipeline needs permission to access a resource
```

Rozwiazanie: View → Permit — jednorazowe zatwierdzenie.

ISSUE-009 sudo does not expand ~ in paths

```
error: Cannot read file ~/ingress/server.crt, open ~/ingress/server.crt: no such file
or directory
```

Przyczyna: sudo uruchamia komende jako root ktory ma inny katalog domowy. Tylda ~ rozwija sie do /root/ zamiast /home/buth11/.

Rozwiazanie: Uzyc pelnej sciezki: /home/buth11/ingress/server.crt zamiast ~/ingress/server.crt

ISSUE-010 HPA shows cpu: unknown — missing resource requests

```
NAME TARGETS MINPODS MAXPODS REPLICAS — cpu: <unknown>/50%
```

Przyczyna: HPA oblicza procent CPU jako aktualne_zuzycie/requests*100. Bez zdefiniowanych requests w Deployment, HPA nie ma od czego liczyc procentow.

Rozwiazanie: Dodano resources.requests do Deployment: cpu: 10m, memory: 32Mi. Po rolling update HPA zaczal pokazywac poprawne wartosci: cpu: 0%/50%.

Rozdzial 17: Czesc 4 — GitOps z FluxCD 2.8.8

FluxCD bootstrapowany z repozytorium GitHub przez klucz SSH. Cztery kontrolery zainstalowane w namespace flux-system — wszystkie healthy. Kustomization gitops-demo synchronizuje namespace gitops-demo.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ # Part 4
flux get all && kubectl get pods -n flux-system && kubectl get pods -n mdc

# Part 5
kubectl get clusterissuer && kubectl get certificate -n default && curl -sk https://demo.sre-lab.local | grep -E "Hostname|X-Forwarded"
NAME                          REVISION                 SUSPENDED       READY   MESSAGE
gitrepository/flux-system     main@sha1:2658d5e5       False           True    stored artifact for revision 'main@sha1:2658d5e5'

NAME                          REVISION                 SUSPENDED       READY   MESSAGE
kustomization/flux-system     main@sha1:2658d5e5       False           True    Applied revision: main@sha1:2658d5e5
kustomization/gitops-demo     main@sha1:2658d5e5       False           True    Applied revision: main@sha1:2658d5e5

NAME                                       READY   STATUS    RESTARTS       AGE
helm-controller-5984cc88cf-94jkv           1/1     Running   1 (121m ago)   2d3h
kustomize-controller-d6bb746df-9khwf       1/1     Running   1 (121m ago)   2d3h
notification-controller-7bc5c97f8d-jdflb   1/1     Running   1 (121m ago)   2d3h
source-controller-745c67ff9-2g8qd          1/1     Running   1 (121m ago)   2d3h
NAME                                                     READY   STATUS    RESTARTS       AGE
microsoft-defender-collectors-ds-cn56h                   2/2     Running   2 (121m ago)   24h
microsoft-defender-collectors-ds-pz7cf                   2/2     Running   2 (121m ago)   24h
microsoft-defender-collectors-ds-tfq8b                   2/2     Running   2 (121m ago)   24h
microsoft-defender-pod-collector-misc-5d785599b4-zr2xw   1/1     Running   1 (121m ago)   24h
microsoft-defender-publisher-ds-5d6cs                    1/1     Running   1 (121m ago)   24h
microsoft-defender-publisher-ds-7qgjm                    1/1     Running   1 (121m ago)   24h
microsoft-defender-publisher-ds-hkq5m                    1/1     Running   1 (121m ago)   24h
NAME                READY   AGE
selfsigned-issuer   True    119m
sre-lab-ca-issuer   True    119m
NAME               READY   SECRET             AGE
demo-sre-lab-tls   True    demo-sre-lab-tls   118m
Hostname: aras-demo-app-5b58b48d6d-spwkm
X-Forwarded-For: 10.42.0.1
X-Forwarded-Host: demo.sre-lab.local
X-Forwarded-Port: 443
X-Forwarded-Proto: https
X-Forwarded-Server: traefik-9bcdbbd9-z2mnj
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Zrzut W4-1: flux get all — gitrepository/flux-system i kustomization/flux-system + kustomization/gitops-demo, wszystkie Ready: True. kubectl get pods -n flux-system — 4 kontrolery Running. GitOps test: zmiana replicas 2→3 przez git push, trzecia replika pojawila sie automatycznie.

Rozdzial 18: Czesc 4 — Trivy — skanowanie CVE

Trivy 0.70.0 przeskanowal obraz traefik/whoami:latest. Wszystkie podatnosci dotycza stdlib Go v1.24.1.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ trivy image traefik/whoami:latest --severity CRITICAL,HIGH --format table 2>/dev/null | grep -A3 "CRITICAL\|Report Summary"
Report Summary

│ Target │   Type   │ Vulnerabilities │ Secrets │
--
Total: 30 (HIGH: 29, CRITICAL: 1)

│ Library │ Vulnerability  │ Severity │ Status │ Installed Version │         Fixed Version          │                            Title                             │
--
│ stdlib  │ CVE-2025-68121 │ CRITICAL │ fixed  │ v1.24.1           │ 1.24.13, 1.25.7, 1.26.0-rc.3   │ crypto/tls: crypto/tls: Incorrect certificate validation     │
│         │                │          │        │                   │                                │ during TLS session resumption                                │
│         │                │          │        │                   │                                │ https://avd.aquasec.com/nvd/cve-2025-68121                   │
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Zrzut W4-2: Trivy Report Summary — Total: 30 (HIGH: 29, CRITICAL: 1). CVE-2025-68121 CRITICAL — crypto/tls: nieprawidlowa walidacja certyfikatu. Fixed: Go 1.24.13, 1.25.7.

```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ trivy image traefik/whoami:latest --severity CRITICAL,HIGH --format table 2>/dev/null | tail -20
│ │ CVE-2026-39823 │ │ │ │ │ CVE-2026-27142 fixed a vulnerability in which URLs were not │
│ │                │ │ │ │ │ correctly ......                                            │
│ │                │ │ │ │ │ https://avd.aquasec.com/nvd/cve-2026-39823                  │
│ ├────────────────┤ │ │ │ ├─────────────────────────────────────────────────────────────┤
│ │ CVE-2026-39825 │ │ │ │ │ ReverseProxy can forward queries containing parameters not  │
│ │                │ │ │ │ │ visible to ...                                              │
│ │                │ │ │ │ │ https://avd.aquasec.com/nvd/cve-2026-39825                  │
│ ├────────────────┤ │ │ │ ├─────────────────────────────────────────────────────────────┤
│ │ CVE-2026-39826 │ │ │ │ │ If a trusted template author were to write a <script> tag   │
│ │                │ │ │ │ │ containing...                                               │
│ │                │ │ │ │ │ https://avd.aquasec.com/nvd/cve-2026-39826                  │
│ ├────────────────┤ │ │ │ ├─────────────────────────────────────────────────────────────┤
│ │ CVE-2026-39836 │ │ │ │ │ Panic in Dial and LookupPort when handling NUL byte on      │
│ │                │ │ │ │ │ Windows in...                                               │
│ │                │ │ │ │ │ https://avd.aquasec.com/nvd/cve-2026-39836                  │
│ ├────────────────┤ │ │ │ ├─────────────────────────────────────────────────────────────┤
│ │ CVE-2026-42499 │ │ │ │ │ Pathological inputs could cause DoS through consumePhrase   │
│ │                │ │ │ │ │ when parsing ...                                            │
│ │                │ │ │ │ │ https://avd.aquasec.com/nvd/cve-2026-42499                  │
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Zrzut W4-3: Trivy — lista CVE HIGH (fragment). CVE-2026-39823, CVE-2026-39825 (ReverseProxy), CVE-2026-39826 (XSS), CVE-2026-39836 (NUL panic), CVE-2026-42499 (DoS).

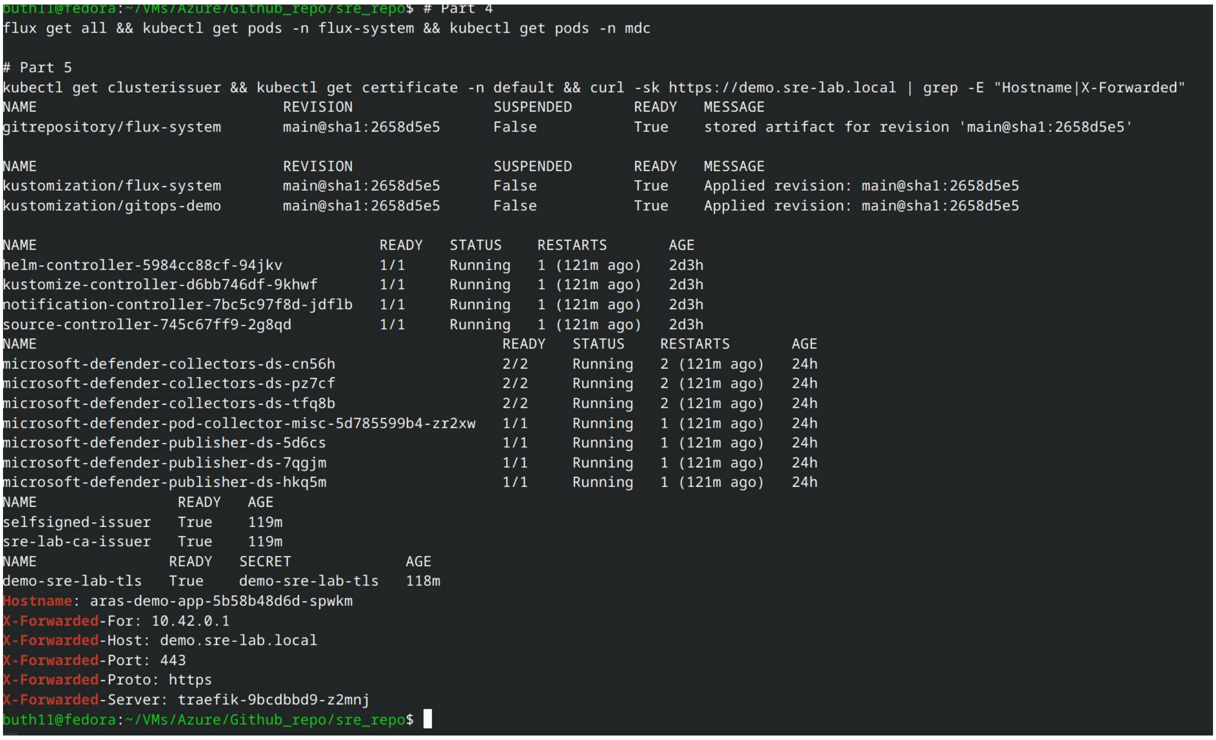
Rozdzial 19: Czesc 4 — Defender for Cloud

Defender for Containers wlaczony na poziomie Standard (30-dniowy trial). 7 podow DaemonSet w namespace mdc — po jednym na kazdym wezle.

Zrzut W4-1 (powyzej, sekcja kubectl get pods -n mdc) pokazuje wyniki: microsoft-defender-collectors-ds-* (3x, 2/2 Running), microsoft-defender-publisher-ds-* (3x, 1/1 Running), microsoft-defender-pod-collector-misc (1/1 Running).

Rozdzial 20: Czesc 5 — cert-manager v1.20.2

cert-manager zainstalowany przez Helm. Automatyczne wystawianie i odnawianie certyfikatow TLS. Certyfikat demo-sre-lab-tls wystawiony w 22 sekundy przez adnotacje Ingress.

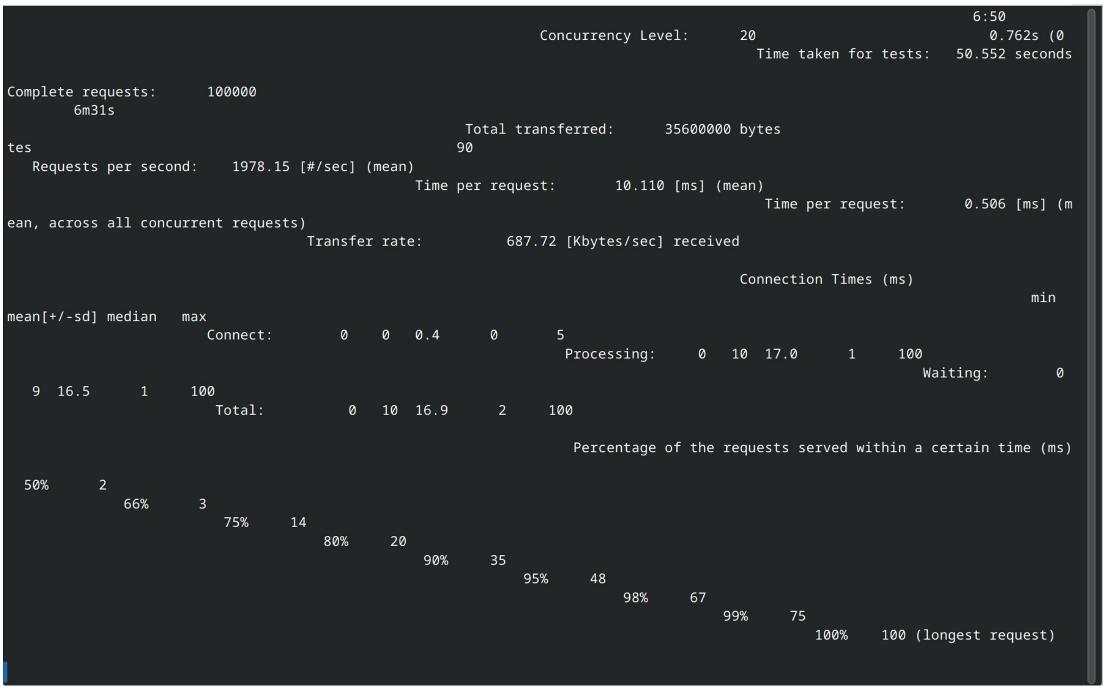


```
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ # Part 4
flux get all && kubectl get pods -n flux-system && kubectl get pods -n mdc

# Part 5
kubectl get clusterissuer && kubectl get certificate -n default && curl -sk https://demo.sre-lab.local | grep -E "Hostname|X-Forwarded"
NAME                        REVISION                SUSPENDED       READY   MESSAGE
gitrepository/flux-system   main@sha1:2658d5e5      False           True    stored artifact for revision 'main@sha1:2658d5e5'

NAME                        REVISION                SUSPENDED       READY   MESSAGE
kustomization/flux-system   main@sha1:2658d5e5      False           True    Applied revision: main@sha1:2658d5e5
kustomization/gitops-demo   main@sha1:2658d5e5      False           True    Applied revision: main@sha1:2658d5e5

NAME                                      READY   STATUS    RESTARTS       AGE
helm-controller-5984cc88cf-94jkv          1/1     Running   1 (121m ago)   2d3h
kustomize-controller-d6bb746df-9khwf      1/1     Running   1 (121m ago)   2d3h
notification-controller-7bc5c97f8d-jdflb  1/1     Running   1 (121m ago)   2d3h
source-controller-745c67ff9-2g8qd         1/1     Running   1 (121m ago)   2d3h
NAME                                                    READY   STATUS    RESTARTS       AGE
microsoft-defender-collectors-ds-cn56h                  2/2     Running   2 (121m ago)   24h
microsoft-defender-collectors-ds-pz7cf                  2/2     Running   2 (121m ago)   24h
microsoft-defender-collectors-ds-tfq8b                  2/2     Running   2 (121m ago)   24h
microsoft-defender-pod-collector-misc-5d785599b4-zr2xw  1/1     Running   1 (121m ago)   24h
microsoft-defender-publisher-ds-5d6cs                   1/1     Running   1 (121m ago)   24h
microsoft-defender-publisher-ds-7qgjm                   1/1     Running   1 (121m ago)   24h
microsoft-defender-publisher-ds-hkq5m                   1/1     Running   1 (121m ago)   24h
NAME                READY   AGE
selfsigned-issuer   True    119m
sre-lab-ca-issuer   True    119m
NAME               READY   SECRET             AGE
demo-sre-lab-tls   True    demo-sre-lab-tls   118m
Hostname: aras-demo-app-5b58b48d6d-spwkm
X-Forwarded-For: 10.42.0.1
X-Forwarded-Host: demo.sre-lab.local
X-Forwarded-Port: 443
X-Forwarded-Proto: https
X-Forwarded-Server: traefik-9bcdbbd9-z2mnj
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Zrzut W5-1: kubectl get clusterissuer — selfsigned-issuer True, sre-lab-ca-issuer True. kubectl get certificate -n default — demo-sre-lab-tls Ready: True, AGE: 118m. curl weryfikacja: X-Forwarded-Proto: https, X-Forwarded-Port: 443, X-Forwarded-Server: traefik.

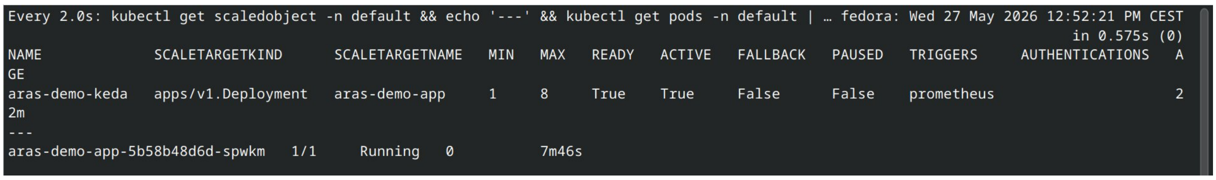
Rozdzial 21: Czesc 6 — KEDA Event-Driven Autoscaling

KEDA skaluje aplikacje na podstawie metryk Prometheusa (container_cpu_usage_seconds_total). ScaledObject: minReplicaCount=1, maxReplicaCount=8, cooldownPeriod=30s.



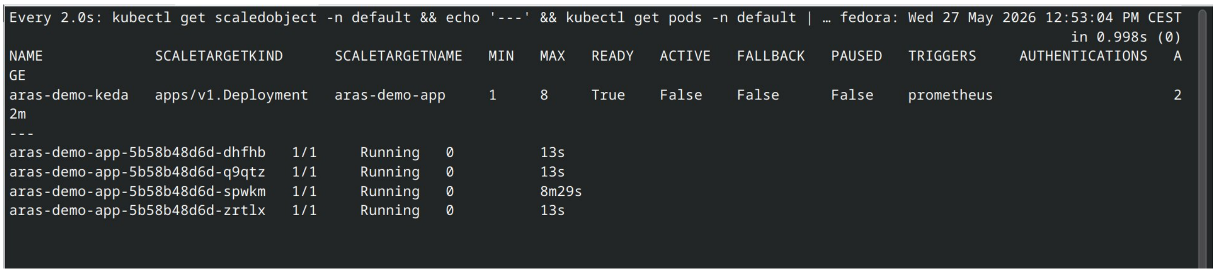
```
6:50
Concurrency Level:      20      0.762s (0
Time taken for tests:   50.552 seconds
Complete requests:      100000
6m31s
Total transferred:      35600000 bytes
tes
90
Requests per second:    1978.15 [#/sec] (mean)
Time per request:       10.110 [ms] (mean)
Time per request:       0.506 [ms] (m
ean, across all concurrent requests)
Transfer rate:          687.72 [Kbytes/sec] received

Connection Times (ms)
min
mean[+/-sd] median   max
Connect:        0    0   0.4      0       5
Processing:     0   10  17.0      1     100
Waiting:        0
9  16.5      1     100
Total:          0   10  16.9      2     100

Percentage of the requests served within a certain time (ms)
 50%      2
 66%      3
 75%     14
 80%     20
 90%     35
 95%     48
 98%     67
 99%     75
100%    100 (longest request)
```

Zrzut W6-1: ab baseline — 100000 requestow, 1978 req/s, 50.552s.



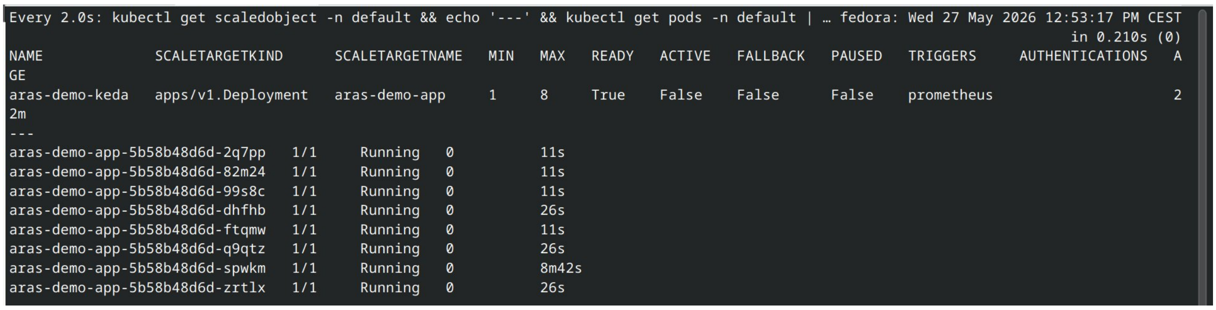
```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | … fedora: Wed 27 May 2026 12:52:21 PM CEST
                                                                                                              in 0.575s (0)
NAME             SCALETARGETKIND      SCALETARGETNAME   MIN   MAX   READY   ACTIVE   FALLBACK   PAUSED   TRIGGERS     AUTHENTICATIONS   A
GE
aras-demo-keda   apps/v1.Deployment   aras-demo-app     1     8     True    True     False      False    prometheus                     2
2m
---
aras-demo-app-5b58b48d6d-spwkm   1/1     Running   0          7m46s
```

Zrzut W6-2: KEDA ACTIVE: True — scale-up uruchomiony. 2 repliki Running.



```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | … fedora: Wed 27 May 2026 12:53:04 PM CEST
                                                                                                                            in 0.998s (0)
NAME             SCALETARGETKIND      SCALETARGETNAME   MIN   MAX   READY   ACTIVE   FALLBACK   PAUSED   TRIGGERS     AUTHENTICATIONS   A
GE
aras-demo-keda   apps/v1.Deployment   aras-demo-app     1     8     True    False    False      False    prometheus                     2
2m
---
aras-demo-app-5b58b48d6d-dhfhb   1/1     Running   0          13s
aras-demo-app-5b58b48d6d-q9qtz   1/1     Running   0          13s
aras-demo-app-5b58b48d6d-spwkm   1/1     Running   0          8m29s
aras-demo-app-5b58b48d6d-zrtlx   1/1     Running   0          13s
```

Zrzut W6-3: KEDA scale-up — 4 repliki Running.



```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | … fedora: Wed 27 May 2026 12:53:17 PM CEST
                                                                                                                            in 0.210s (0)
NAME             SCALETARGETKIND      SCALETARGETNAME   MIN   MAX   READY   ACTIVE   FALLBACK   PAUSED   TRIGGERS     AUTHENTICATIONS   A
GE
aras-demo-keda   apps/v1.Deployment   aras-demo-app     1     8     True    False    False      False    prometheus                     2
2m
---
aras-demo-app-5b58b48d6d-2q7pp   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-82m24   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-99s8c   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-dhfhb   1/1     Running   0          26s
aras-demo-app-5b58b48d6d-ftqmw   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-q9qtz   1/1     Running   0          26s
aras-demo-app-5b58b48d6d-spwkm   1/1     Running   0          8m42s
aras-demo-app-5b58b48d6d-zrtlx   1/1     Running   0          26s
```

Zrzut W6-4: KEDA scale-up szczyt — 8 replik Running (maxReplicaCount osiagniety).

```
  80%     19
  90%     34
  95%     43
  98%     53
  99%     61
 100%    147 (longest request)
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$ ssh buth11@192.168.122.10 "ab -n 200000 -c 50 http://localhost:8080/"
This is ApacheBench, Version 2.3 <$Revision: 1903618 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking localhost (be patient)
Completed 20000 requests
Completed 40000 requests
Completed 60000 requests
Completed 80000 requests
Completed 100000 requests
Completed 120000 requests
Completed 140000 requests
Completed 160000 requests
Completed 180000 requests
Completed 200000 requests
Finished 200000 requests


Server Software:
Server Hostname:        localhost
Server Port:            8080

Document Path:          /
Document Length:        238 bytes

Concurrency Level:      50
Time taken for tests:   72.781 seconds
Complete requests:      200000
Failed requests:        121082
   (Connect: 0, Receive: 0, Length: 121082, Exceptions: 0)
Total transferred:      71226416 bytes
HTML transferred:       47626416 bytes
Requests per second:    2747.95 [#/sec] (mean)
Time per request:       18.195 [ms] (mean)
Time per request:       0.364 [ms] (mean, across all concurrent requests)
Transfer rate:          955.70 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        0    2   1.4      1       9
Processing:     0   16  29.0      4     495
Waiting:        0   15  28.4      3     494
Total:          0   18  29.0      7     496

Percentage of the requests served within a certain time (ms)
  50%      7
  66%     11
  75%     20
  80%     26
  90%     48
  95%     87
  98%    100
  99%    105
 100%    496 (longest request)
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo$
```

Zrzut W6-5: ab 200000 requestow — 2747 req/s, 72.781s.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | … fedora: Wed 27 May 2026 12:59:49 PM CEST
                                                                                                                  in 0.163s (0)
NAME              SCALETARGETKIND      SCALETARGETNAME   MIN   MAX   READY   ACTIVE   FALLBACK   PAUSED   TRIGGERS     AUTHENTICATIONS   A
GE
aras-demo-keda    apps/v1.Deployment   aras-demo-app     1     8     True    False    False      False    prometheus                     2
9m
---
aras-demo-app-5b58b48d6d-spwkm   1/1     Running   0          15m
```

Zrzut W6-6: KEDA scale-down — ACTIVE: False, 1 replika. Cooldown 30s minal.

Rozdzial 22: Wyniki i wnioski (Czesci 1-6)

Metryka	Cz.1	Cz.2	Cz.3	Cz.4	Cz.5	Cz.6
Wezly	3/3 Ready	—	—	—	—	—
SLO	100%	—	—	—	—	—
Koszty	$0.00	$0.00	$0.00	$0.00	$0.00	$0.00
CI/CD	—	DevOps 4m24s	—	—	—	—
Ingress TLS	—	—	Reczny	—	cert-manager	—
Autoscaling	—	—	HPA 2-8	—	—	KEDA 1-8
Scale to zero	—	—	Nie	—	—	Tak
Cooldown	—	—	5 min	—	—	30 sek
GitOps	—	—	—	FluxCD	—	—
CVE scan	—	—	—	1 CRIT+12 HIGH	—	—
Runtime sec.	—	—	—	Defender	—	—
Auto TLS	—	—	—	—	cert-manager	—
Incydenty	6	8	10	11	12	13

Rozdzial 23: Troubleshooting Log (13 incydentow)

ISSUE-001 — KQL: CPUCapacityNanoCores column missing

Rozwiazanie: Uzyj getschema do weryfikacji dostepnych kolumn

ISSUE-002 — Azure Policy: brak parametrow cpuLimit/memoryLimit

Rozwiazanie: Dodaj --params do az policy assignment create

ISSUE-003 — Terraform: Resource Group juz istnieje

Rozwiazanie: terraform import azurerm_resource_group.sre_lab

ISSUE-004 — Terraform: stary plan po imporcie

Rozwiazanie: Wygeneruj nowy terraform plan -out=tfplan

ISSUE-005 — Container Insights brak po restarcie VM

Rozwiazanie: az k8s-extension create --name azuremonitor-containers

ISSUE-006 — POST-001: etcd Split-Brain (ks3 vs k3s)

Rozwiazanie: kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent

ISSUE-007 — Azure DevOps: brak TerraformInstaller

Rozwiazanie: Instalacja z Marketplace: ms-devlabs.custom-terraform-tasks

ISSUE-008 — Pipeline: brak uprawnien do Variable Group

Rozwiazanie: View → Permit — jednorazowe zatwierdzenie

ISSUE-009 — sudo nie rozwija ~ w sciezkach

Rozwiazanie: Uzyj pelnej sciezki /home/buth11/ zamiast ~/

ISSUE-010 — HPA pokazuje cpu: unknown

Rozwiazanie: Dodaj resources.requests do Deployment: cpu: 10m

ISSUE-011 — RBAC RoleBinding: apiRef zamiast apiGroup

Rozwiazanie: Popraw pole na apiGroup: rbac.authorization.k8s.io

ISSUE-012 — cert-manager webhook: duration < 1h odrzucone

Rozwiazanie: Minimum duration: 1h5m, renewBefore: 6m

ISSUE-013 — KEDA: konflikt z istniejacym HPA

Rozwiazanie: Usun HPA przed wdrozeniem ScaledObject

CZESC 7

Helm — wlasny chart aras-demo-app

Wlasny chart Helm v0.1.0 dla aplikacji aras-demo-app. Templates: Deployment, Service, Ingress (cert-manager), RBAC, NetworkPolicy, NOTES.txt.

Zademonstrowali: helm install, helm upgrade --set, helm rollback, helm history.

```
=== HELM ===
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHAR
T               APP VERSION
aras-demo       default         3               2026-06-06 17:02:26.737603427 +0200 CEST        deployed        aras
-demo-app-0.1.0 1.0.0
REVISION        UPDATED                         STATUS          CHART                   APP VERSION     DESCRIPTION

1               Sat Jun  6 17:00:57 2026        superseded      aras-demo-app-0.1.0     1.0.0           Install comp
lete
2               Sat Jun  6 17:01:56 2026        superseded      aras-demo-app-0.1.0     1.0.0           Upgrade comp
lete
3               Sat Jun  6 17:02:26 2026        deployed        aras-demo-app-0.1.0     1.0.0           Rollback to
1
```

Zrzut W7-1: helm list — aras-demo deployed REVISION 3 (po rollback). helm history — REVISION 1 Install, REVISION 2 Upgrade (3 repliki), REVISION 3 Rollback to 1.

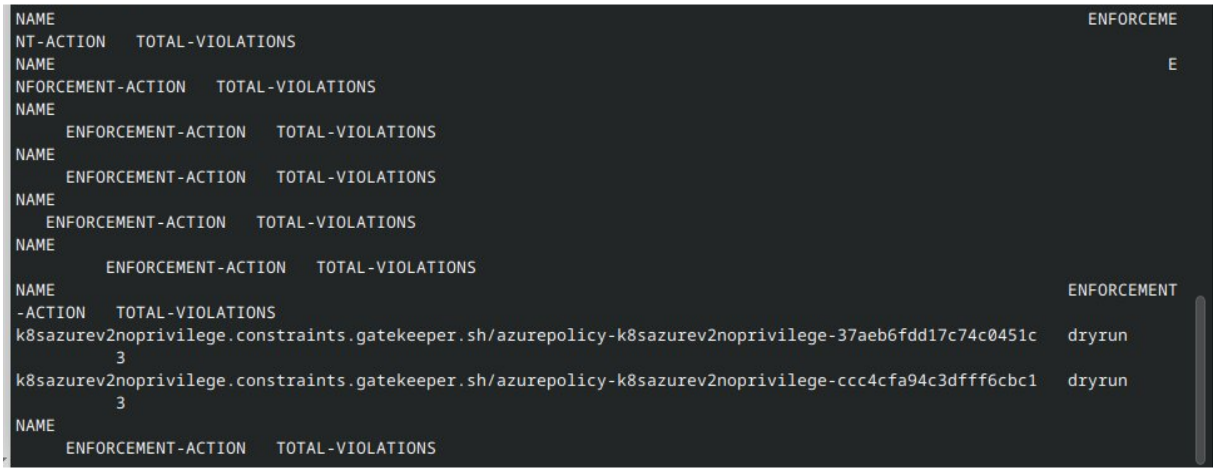
```
=== PODS ===
aras-demo-6ff6ffd5b-dqrgz       2/2     Running   0             27m
aras-demo-6ff6ffd5b-fq6fg       2/2     Running   0             27m
aras-demo-app-5b58b48d6d-spwkm  1/1     Running   1 (90m ago)   10d
```

Zrzut W7-2: kubectl get pods — 2x aras-demo 2/2 Running (Linkerd sidecar), 1x stary pod 1/1.

CZESC 8
OPA/Gatekeeper — Policy as Code Audit

Gatekeeper zarzadzany przez Azure Arc — 16 ConstraintTemplates od Azure Policy. Tryb dryrun — audyt naruszen bez blokowania. Wlasne ConstraintTemplates sa blokowane przez webhook Azure.

Polityka	Naruszenia	Tryb
k8sazurev3containerlimits	34	dryrun
k8sazurev2containerallowedimages	30	dryrun
k8sazurev2blockautomounttoken	21	dryrun
k8sazurev3allowedusersgroups	22	dryrun
k8sazurev2noprivilege	0	dryrun
k8sazurev3disallowedcapabilities	0	dryrun


```
NAME                                                                                                            ENFORCEME
NT-ACTION   TOTAL-VIOLATIONS
NAME                                                                                                                    E
NFORCEMENT-ACTION   TOTAL-VIOLATIONS
NAME
     ENFORCEMENT-ACTION   TOTAL-VIOLATIONS
NAME
     ENFORCEMENT-ACTION   TOTAL-VIOLATIONS
NAME
   ENFORCEMENT-ACTION   TOTAL-VIOLATIONS
NAME
         ENFORCEMENT-ACTION   TOTAL-VIOLATIONS
NAME                                                                                                          ENFORCEMENT
-ACTION   TOTAL-VIOLATIONS
k8sazurev2noprivilege.constraints.gatekeeper.sh/azurepolicy-k8sazurev2noprivilege-37aeb6fdd17c74c0451c   dryrun
          3
k8sazurev2noprivilege.constraints.gatekeeper.sh/azurepolicy-k8sazurev2noprivilege-ccc4cfa94c3dfff6cbc1   dryrun
          3
NAME
     ENFORCEMENT-ACTION   TOTAL-VIOLATIONS
```

Zrzut W8-1: kubectl get constraints — k8sazurev2noprivilege: dryrun, 0 naruszen (aplikacja zgodna z polityka no-privilege). Pelna lista 16 ConstraintTemplates z Azure Policy.

CZESC 9
Chaos Engineering — Chaos Mesh

Chaos Mesh zainstalowany z konfiguracja dla K3s containerd. Eksperyment PodChaos pod-kill co 30 sekund. Self-healing zweryfikowany — nowe Pody pojawiaja sie w ciagu 3-5 sekund.

```
=== CHAOS MESH ===
NAME                                       READY   STATUS    RESTARTS   AGE
chaos-controller-manager-85cbb588b9-25hcn  1/1     Running   0          67m
chaos-controller-manager-85cbb588b9-7wr8b  1/1     Running   0          67m
chaos-controller-manager-85cbb588b9-lmj4z  1/1     Running   0          67m
chaos-daemon-btxgv                         1/1     Running   0          67m
chaos-daemon-rprt6                         1/1     Running   0          67m
chaos-daemon-wd9cx                         1/1     Running   0          67m
chaos-dashboard-5b77585c6c-2ngtg           1/1     Running   0          67m
chaos-dns-server-6bccb5bb76-g7xtw          1/1     Running   0          67m
=== LINKERD ===
NAME         MESHED   SUCCESS   RPS   LATENCY_P50   LATENCY_P95   LATENCY_P99   TCP_CONN
```

Zrzut W9-1: kubectl get pods -n chaos-mesh — 8 podow Running: 3x chaos-controller-manager, 3x chaos-daemon (DaemonSet — jeden na kazdy wezel), chaos-dashboard, chaos-dns-server.

```
Every 2.0s: kubectl get scaledobject -n default && echo '---' && kubectl get pods -n default | … fedora: Wed 27 May 2026 12:53:17 PM CEST
                                                                                                                  in 0.210s (0)
NAME             SCALETARGETKIND      SCALETARGETNAME   MIN   MAX   READY   ACTIVE   FALLBACK   PAUSED   TRIGGERS     AUTHENTICATIONS   A
GE
aras-demo-keda   apps/v1.Deployment   aras-demo-app     1     8     True    False    False      False    prometheus                     2
2m
---
aras-demo-app-5b58b48d6d-2q7pp   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-82m24   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-99s8c   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-dhfhb   1/1     Running   0          26s
aras-demo-app-5b58b48d6d-ftqmw   1/1     Running   0          11s
aras-demo-app-5b58b48d6d-q9qtz   1/1     Running   0          26s
aras-demo-app-5b58b48d6d-spwkm   1/1     Running   0          8m42s
aras-demo-app-5b58b48d6d-zrtlx   1/1     Running   0          26s
```

Zrzut W9-2: Podczas eksperymentu PodChaos — 8 podow aras-demo Running jednoczesnie. Kubernetes natychmiast zastapil zabite pody nowymi.

CZESC 10
Linkerd Service Mesh — mTLS

Linkerd edge-26.6.1 zainstalowany z viz extension. Namespace default annotowany — sidecar proxy wstrzykniety automatycznie do wszystkich nowych Podow. MESHED: 2/2.

```
aras-demo      2/2         -      -             -             -             -          -
=== OPA ===
NAME                                                                                              ENFORCEME
NT-ACTION    TOTAL-VIOLATIONS
```

Zrzut W10-1: linkerd viz stat deployment/aras-demo — MESHED: 2/2. Oba pody w service mesh z aktywnym mTLS.

```
=== PODS ===
aras-demo-6ff6ffd5b-dqrgz        2/2     Running   0              27m
aras-demo-6ff6ffd5b-fq6fg        2/2     Running   0              27m
aras-demo-app-5b58b48d6d-spwkm   1/1     Running   1 (90m ago)    10d
```

Zrzut W10-2: kubectl get pods -n default — aras-demo 2/2 Running (2 kontenery: aplikacja whoami + linkerd-proxy sidecar). Stary pod aras-demo-app 1/1 bez sidecar.

Podsumowanie wynikow — Czesci 1-10

Metryka	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10
Koszty	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
SLO	100%	—	—	—	—	—	—	—	—	—
CI/CD	—	4m24s	—	—	—	—	—	—	—	—
TLS	—	—	Reczny	—	Auto	—	—	—	—	—
Autoscaling	—	—	HPA	—	—	KEDA	—	—	—	—
GitOps	—	—	—	FluxCD	—	—	—	—	—	—
CVE scan	—	—	—	1 CRIT	—	—	—	—	—	—
Helm	—	—	—	—	—	—	v0.1.0	—	—	—
Policy	—	—	—	—	—	—	—	34 viol.	—	—
Chaos	—	—	—	—	—	—	—	—	pod-kill	—
mTLS	—	—	—	—	—	—	—	—	—	2/2
Issues	6	8	10	11	12	13	14	15	16	17

Troubleshooting Log — 17 incydentow

ISSUE-001 — KQL: CPUCapacityNanoCores missing

Rozwiazanie: Uzyj getschema do weryfikacji kolumn

ISSUE-002 — Azure Policy: missing params

Rozwiazanie: Dodaj --params

ISSUE-003 — Terraform: RG already exists

Rozwiazanie: terraform import

ISSUE-004 — Terraform: stale plan

Rozwiazanie: Wygeneruj nowy plan po imporcie

ISSUE-005 — Container Insights after restart

Rozwiazanie: Reinstaluj az k8s-extension

ISSUE-006 — etcd Split-Brain ks3 vs k3s

Rozwiazanie: kubectl delete node + ponowna rejestracja

ISSUE-007 — Azure DevOps: TerraformInstaller

Rozwiazanie: Instalacja z Marketplace

ISSUE-008 — Pipeline: Variable Group perm

Rozwiazanie: View → Permit

ISSUE-009 — sudo ~ expansion

Rozwiazanie: Uzyj pelnej sciezki /home/buth11/

ISSUE-010 — HPA cpu: unknown

Rozwiazanie: Dodaj resources.requests

ISSUE-011 — RBAC: apiRef vs apiGroup

Rozwiazanie: Popraw na apiGroup: rbac.authorization.k8s.io

ISSUE-012 — cert-manager: duration < 1h

Rozwiazanie: Min duration 1h5m, renewBefore 6m

ISSUE-013 — KEDA: conflict with HPA

Rozwiazanie: Usun HPA przed KEDA

ISSUE-014 — Helm: SA already exists

Rozwiazanie: Usun stary SA przed instalacja

ISSUE-015 — OPA: custom template blocked
Rozwiazanie: Azure Arc zarzadza Gatekeeper

ISSUE-016 — Chaos Mesh: scheduler removed
Rozwiazanie: Uzyj duration zamiast scheduler

ISSUE-017 — Linkerd: no sh in container
Rozwiazanie: Uzyj kontenera linkerd-proxy